

Lecture Notes on Elliptic curves

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Remark 0.1. These are lecture notes based on the course elliptic curves at Humboldt University Berlin SoSe2026 taught by Jiaming Chen. In the later parts I follow Silverman very closely, but rephrase certain parts to fit within my understanding. Any typos and mistakes are to be strictly attributed to myself.

§1 Lecture 1

The first object we discuss in this course are elliptic functions. To discuss these we recall a few basic definitions and terminology.

Definition 1.1. A **Lattice** $\Lambda \subseteq \mathbb{C}$ is a discrete subgroup which generates $\mathbb{C}$ as a $\mathbb{R}$ vector space.

Remark 1.2. One easily verifies that any lattice admits a $\mathbb{Z}$ basis, i.e. $\Lambda = a\mathbb{Z} \oplus b\mathbb{Z}$, for $a, b \in \mathbb{C}$. This can be seen by choosing a $\mathbb{R}$ basis of $\mathbb{C}$ contained in Λ with minimal determinant.

Definition 1.3. A **fundamental domain** for a lattice is a subset $D \subset \mathbb{C}$ of the form:

$$D = \{a + t_2\omega_1 + t_1\omega_2 \mid 0 \leq t_1, t_2 < 1\},$$

where $a \in \mathbb{C}$, ω_1, ω_2 is a $\mathbb{Z}$ -basis of Λ and $a \in \mathbb{C}$.

Remark 1.4. In the literature, one sometimes sees a slightly more general definition, namely a measurable subset of $\mathbb{C}$ which consists of a set of representatives of $\mathbb{C}/\Lambda$.

Definition 1.5. Let $\Lambda \subseteq \mathbb{C}$ be a lattice, an **elliptic function** with respect to Λ , is a meromorphic function

$$f: \mathbb{C} \rightarrow \mathbb{P}\mathbb{C}^1,$$

which is Λ periodic.

Remark 1.6. If we invoke the theory of Riemann surfaces, we may alternatively view an elliptic function as a meromorphic function on the genus one surface $\mathbb{C}/\Lambda$.

We will denote by $\mathbb{C}(\Lambda)$ the set of elliptic functions on $\mathbb{C}$ with respect to Λ . It carries a natural field structure.

One should notice that only requiring our functions to be meromorphic rather than entire is the correct choice to make. Indeed, entire functions which are Λ -periodic are bounded and hence constant by Liouville's theorem. In fact, we can already say slightly more about elliptic functions:

Theorem 1.7

Let $\Lambda \subseteq \mathbb{C}$ be a lattice, $D \subseteq \mathbb{C}$ a fundamental domain and $f \in \mathbb{C}(\Lambda)^\times$. Then the following three equations hold:

$$\sum_{x \in D} \text{res}_x(f) = 0 \tag{1}$$

$$\sum_{x \in D} \text{ord}_x(f) = 0 \tag{2}$$

$$\sum_{x \in D} \text{ord}_x(f)x \in \Lambda. \tag{3}$$

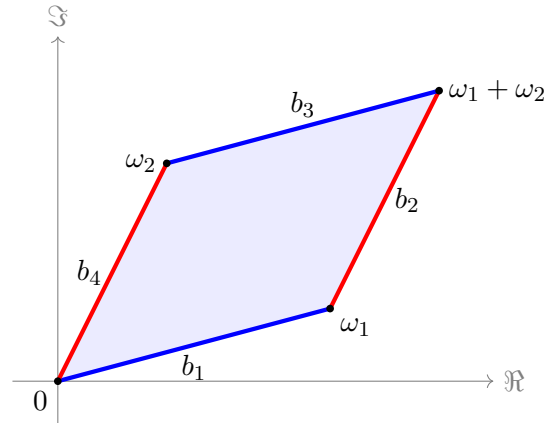
Remark 1.8. Notice that the above sums are actually finite sums since D is bounded. Zeros/poles of a meromorphic function are discrete.

Proof. notice that the above equations do not depend on choice of fundamental domain as they may be written as equations in $\mathbb{C}/\Gamma$. Thus, we may assume D is such that none

of the poles or singularities lie on ∂D (if not slightly perturb it). Let us now prove (1). Using the residue theorem, we may write:

$$\sum_{x \in D} \text{res}_x(f) = \frac{1}{2\pi i} \int_{\partial D} f(z) dz.$$

split the boundary ∂D into four parts, corresponding to the figure below. These terms pairwise cancel out by the Λ -periodicity of f .



Let us now prove part (2), and let us assume $\text{ord}_x(f) = n \in \mathbb{Z}$. Then, locally near x , we may write $f(z) = (z - a)^n h(z)$ where h is holomorphic and nonzero near x . In the neighborhood where this holds we find

$$\frac{f'(z)}{f(z)} = \frac{n(z - x)^{n-1} h(z) + (z - x)^n h'(z)}{(z - x)^n h(z)} = \frac{n}{z - x} + \frac{h'(z)}{h(z)}.$$

where evidently the last expression has residue n at x as the second term is holomorphic. Thus we conclude

$$\text{ord}_x(f) = \text{res}_x \left(\frac{f'}{f} \right).$$

Assuming f is non constant (otherwise the statement is trivial), the function $\frac{f'}{f}$ is nonzero and hence the statement follows from (1). The statement of (3) is also a consequence of the residue theorem, once again let us write locally:

$$f(z) = (z - x)^n h(z),$$

with the same conditions on h as above. Locally we have

$$z \cdot \frac{f'(z)}{f(z)} = \frac{z \cdot n}{z - x} + z \frac{h'(z)}{h(z)}.$$

The residue of the right side at x , is $x \cdot n$ and thus:

$$\sum_{x \in D} \text{ord}_x(f) x = \sum_{x \in D,} \text{res}_x \left(z \cdot \frac{f'}{f} \right)$$

Using the residue theorem, the last sum evaluates to:

$$\sum_{x \in D,} \text{res} \left(z \cdot \frac{f'}{f} \right) = \frac{1}{2\pi i} \int_{\partial D} z \cdot \frac{f'(z)}{f(z)} dz.$$

According to the figure above, decompose the boundary ∂D into the four parts b_1, b_2, b_3, b_4 and rewrite the integral as

$$\begin{aligned} \frac{1}{2\pi i} \int_{\partial D} z \frac{f'(z)}{f(z)} dz &= \frac{1}{2\pi i} \sum_{j=1}^4 \int_{b_j} z \frac{f'(z)}{f(z)} dz \\ &= \frac{1}{2\pi i} \left(\int_{b_1} z \frac{f'(z)}{f(z)} dz - \int_{b_1} (z + \omega_2) \frac{f'(z)}{f(z)} dz \right) \\ &\quad + \frac{1}{2\pi i} \left(\int_{b_2} z \frac{f'(z)}{f(z)} dz - \int_{b_2} (z - \omega_1) \frac{f'(z)}{f(z)} dz \right) \\ &= \omega_1 \frac{1}{2\pi i} \int_{b_2} \frac{f'(z)}{f(z)} dz - \omega_2 \frac{1}{2\pi i} \int_{b_1} \frac{f'(z)}{f(z)} dz. \end{aligned}$$

Thus, it suffices to prove that

$$\frac{1}{2\pi i} \int_{b_i} \frac{f'(z)}{f(z)} dz \in \mathbb{Z}.$$

We will do this for $i = 1$, the other cases are analogous. Consider the path

$$\gamma : [0, 1] \rightarrow \mathbb{C}^\times, \quad t \mapsto f(a + t\omega_1)$$

This path is well defined and nonzero since we have assumed none of the poles nor zeros to be on the boundary of ∂D . Furthermore it is closed as f is Λ -periodic. Thus, by the winding number theorem from complex analysis we have

$$\frac{1}{2\pi i} \int_{b_1} \frac{f'(z)}{f(z)} dz = \frac{1}{2\pi i} \int_{\gamma} \frac{1}{z} dz = \text{ind}(\gamma) \in \mathbb{Z}.$$

□

Corollary 1.9

A non constant elliptic function has atleast two poles when counted with multiplicity.

Proof. $f \in \mathbb{C}(\Lambda)$ having only one simple pole contradicts (1) in Theorem 1.7

□

§2 Lecture 2

§2.1 The Weierstraß $\wp$ function

Definition 2.1. Let $\Lambda \subseteq \mathbb{C}$ be a lattice, we define

$$\begin{aligned} G_{2k}(\Lambda) &:= \sum_{\omega \in \Lambda \setminus \{0\}} \frac{1}{\omega^{2k}} \\ \wp_{\Lambda}(z) &:= \frac{1}{z^2} + \sum_{\omega \in \Lambda \setminus \{0\}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right). \end{aligned}$$

G_{2k} is called the **Eisenstein series** for Λ and $\wp_{\Lambda}(z)$ is called the **Weierstraß $\wp$ function** for Λ .

Proposition 2.2

We have the following convergence properties for G_{2k} and $\wp$

1. $G_{2k}(\Lambda)$ converges absolutely for all $k > 1$.
2. $\wp_\Lambda(z)$ converges absolutely for any $z \in \mathbb{C} \setminus \Lambda$ and uniformly on any compact subset of $\mathbb{C} \setminus \Lambda$.
3. $\wp_\Lambda$ is an even elliptic function with double pole and residue 0 at any lattice point and no other poles.

Proof. 1. Let $k > 1$ and compute :

$$\sum_{\omega \in \Lambda \setminus \{0\}} \left| \frac{1}{\omega^{2k}} \right| = \sum_{\substack{\omega \in \Lambda \\ 0 < |\omega| < 1}} \left| \frac{1}{\omega^{2k}} \right| + \sum_{\substack{\omega \in \Lambda \\ |\omega| \geq 1}} \left| \frac{1}{\omega^{2k}} \right|$$

The first of the two sums is a finite sum and thus finite. For the second sum we have:

$$\begin{aligned} \sum_{\substack{\omega \in \Lambda \\ |\omega| \geq 1}} \left| \frac{1}{\omega^{2k}} \right| &= \sum_{N=1}^{\infty} \sum_{\substack{\omega \in \Lambda \setminus \{0\} \\ N \leq |\omega| < N+1}} \left| \frac{1}{\omega^{2k}} \right| \\ &\leq \sum_{N=1}^{\infty} \frac{|\{\omega \in \Lambda \mid N \leq |\omega| < N+1\}|}{N^{2k}}. \end{aligned}$$

Thus, to conclude it suffices to prove that

$$|\{\omega \in \Lambda \mid N \leq |\omega| < N+1\}| \in O(N),$$

For if this were true we could have

$$\sum_{N=1}^{\infty} \frac{|\{\omega \in \Lambda \mid N \leq |\omega| < N+1\}|}{N^{2k}} \leq \sum_{N=1}^{\infty} \frac{CN}{N^{2k}} = C \sum_{N=1}^{\infty} \frac{1}{N^{2k-1}} < \infty$$

since $2k - 1 > 1$. We will assume this estimate for now, and prove it in the exercises.

2. Let us first show absolute convergence. Let $z \in \mathbb{C} \setminus \Lambda$ be a fixed complex number. We investigate the second summand in the definition of the Weierstraß function. We have:

$$\begin{aligned} \sum_{\omega \in \Lambda \setminus \{0\}} \left| \frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right| &= \sum_{\omega \in \Lambda \setminus \{0\}} \left| \frac{\omega^2 - (z - \omega)^2}{\omega^2(z - \omega)^2} \right| \\ &= \sum_{\omega \in \Lambda \setminus \{0\}} \left| \frac{z(z - 2\omega)}{\omega^2(z - \omega)^2} \right| \end{aligned}$$

to show that the last sum converges we can estimate:

$$\left| \frac{z(z - 2\omega)}{\omega^2(z - \omega)^2} \right| = |z| \left| \frac{z - 2\omega}{\omega^2(z - \omega)^2} \right| \leq |z| \frac{|z| + |2z - 2\omega|}{|\omega^2(z - \omega)^2|}$$

For large enough ω , we have $|z| \leq |z - \omega|$, and thus, for large ω we can estimate:

$$|z| \frac{|z| + |2z - 2\omega|}{|\omega^2(z - \omega)^2|} \leq \frac{3|z|}{|\omega^2(z - \omega)|}.$$

Furthermore, by the reverse triangle inequality we have $|z - \omega| \geq |\omega| - |z|$ and again choosing ω sufficiently large we find $|\omega| - |z| \geq \frac{1}{2}|\omega|$. This allows us to estimate for large ω :

$$\frac{3|z|}{|\omega^2(z - \omega)|} \leq \frac{6|z|}{\omega^3}.$$

Now fix a constant $C > 0$ (dependent only on z) such that all these estimates hold whenever $|\omega| > C$ and conclude:

$$\begin{aligned} \sum_{\omega \in \Lambda \setminus \{0\}} \left| \frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right| &= \sum_{\substack{\omega \in \Lambda \setminus \{0\} \\ |\omega| < C}} \left| \frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right| + \sum_{\substack{\omega \in \Lambda \setminus \{0\} \\ |\omega| \geq C}} \left| \frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right| \\ &\leq \sum_{\substack{\omega \in \Lambda \setminus \{0\} \\ |\omega| < C}} \left| \frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right| + \sum_{\substack{\omega \in \Lambda \setminus \{0\} \\ |\omega| \geq C}} \frac{6|z|}{|\omega^3|}. \end{aligned}$$

The first sum is a finite sum and the second is finite by 1. Now suppose $K \subseteq \mathbb{C} \setminus \Lambda$ is a compact set and let us denote by $\wp_n(z)$ the partial sum

$$\wp_n(z) = \frac{1}{z^2} + \sum_{\substack{\omega \in \Lambda \\ 0 < |\omega| < n}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right)$$

for two natural numbers $n < m$ we may compute on K :

$$\begin{aligned} \|\wp_n - \wp_m\|_\infty &= \left\| \sum_{\substack{\omega \in \Lambda \\ n \leq |\omega| < m}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right) \right\|_\infty \\ &\leq \left\| \sum_{\substack{\omega \in \Lambda \\ n \leq |\omega|}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right) \right\|_\infty. \end{aligned}$$

If we choose $n > C$ the estimate from the absolute convergence part shows

$$\left\| \sum_{\substack{\omega \in \Lambda \\ n \leq |\omega|}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right) \right\|_\infty \leq \|z\|_\infty \cdot \sum_{\substack{\omega \in \Lambda \setminus \{0\} \\ |\omega| \geq C}} \frac{6}{|\omega^3|} \rightarrow 0$$

Where we receive the last statement from the fact that K is compact (in fact bounded suffices)

3. For any compact set $K \subseteq \mathbb{C} \setminus \Lambda$, Weierstraß $\wp$ is a uniform limit of holomorphic functions and thus holomorphic. Thus the singularities of $\wp(z)$ are isolated and is a meromorphic function. The only possible poles are lattice points. Indeed these

are all poles of order two: for example $\wp(z) \cdot z^2$ is holomorphic near 0 but $\wp(z) \cdot z$ is not. The following calculation also shows that $\wp(z)$ is an even function:

$$\wp(-z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda \setminus \{0\}} \left(\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right) = \wp(z)$$

as the above sum is a reordering of the original definition of $\wp(z)$ which equates to the same as we have absolute convergence and may reorder terms. Lastly, we will show that $\wp(z)$ is elliptic, i.e., Λ -periodic. We will show this by considering derivatives of $\wp(z)$. First note that we may differentiate $\wp$ term by term by 2. Hence if $\omega_0 \in \Lambda$

$$\wp'(z) = -2 \sum_{\omega \in \Lambda} \frac{1}{(z - \omega)^3} = \wp'(z + \omega_0).$$

Integrating this relation we have:

$$\wp(z + \omega_0) = \wp(z) + c(\omega_0),$$

for some constant $c(\omega_0) \in \mathbb{C}$. Our goal is to show that it is zero. Let us choose a $\mathbb{Z}$ basis of Λ : $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$. Then, it suffices to prove that $c(\omega_1) = c(\omega_2) = 0$ since any other shift is a $\mathbb{Z}$ -combination of these shifts. For ω_1 we have:

$$\wp\left(\frac{-\omega_1}{2} + \omega_1\right) = \wp\left(-\frac{\omega_1}{2}\right) + c(\omega_1),$$

which implies that $c(\omega_1) = 0$ since $\wp(z) = \wp(-z)$. The same argument shows the result for ω_2 as well. □

Remark 2.3. in 2, we only used that z is bounded to infer uniform convergence, so we have shown that the Weierstrass function actually converges uniformly on bounded sets away from Λ .

The above proposition gives us our first example of an elliptic function and as it turns out, all others may be generated from these:

Proposition 2.4

Every elliptic function is a rational function in the two variables $\wp(z)$ and $\wp'(z)$. In formulas this means:

$$\mathbb{C}(\Lambda) = \mathbb{C}(\wp(z), \wp'(z))$$

Proof. We prove both inclusions. First it is obvious that for any two polynomials $P, Q \in \mathbb{C}[X, Y]$, with $Q(\wp, \wp') \neq 0$, we have

$$\frac{P(\wp(x), \wp'(x))}{Q(\wp(x), \wp'(x))} \in \mathbb{C}(\Lambda),$$

since both the numerator and denominator are elliptic. Thus it remains to prove the reverse inclusion. Suppose $f(z) \in \mathbb{C}(\Lambda)$, then we may decompose f into an even and an odd part:

$$f(z) = \frac{f(z) - f(-z)}{2} + \frac{f(z) + f(-z)}{2},$$

which are both still elliptic functions. In particular it suffices to prove that all even and all odd elliptic functions are in $\mathbb{C}(\wp, \wp')$. The statement for odd functions follows from the statement for even functions since if f is an odd function, $f\wp'$ is even. Thus let us fix an even elliptic function $f \in \mathbb{C}(\Lambda)$. The first thing to note is that since f is even, we have

$$\text{ord}_x(f) = \text{ord}_{-x}(f) \quad \forall x \in \mathbb{C} \setminus \Lambda.$$

We claim that if $x \in \frac{1}{2}\Lambda$, then $\text{ord}_x(f)$ is even. Indeed, suppose f is even and (wlog) that $\text{ord}_x(f) > 0$ (otherwise consider $\frac{1}{f}$ which is also an even elliptic function). In this case we have:

$$f^{(i)}(x) = (-1)^i f^{(i)}(-x) = (-1)^i f^{(i)}(-x + 2x) = (-1)^i f^{(i)}(x).$$

This forces, in particular, $f^{(i)}(x) = 0$ when i is odd and thus $\text{ord}_x(f) \in 2\mathbb{Z}$. With this in mind, let D be a fundamental domain and choose $H \subseteq D$ such that for all $\omega \in \mathbb{C}$, exactly one of the two elements $\{\omega, -\omega\}$ is represented by an element in H in the quotient $\mathbb{C}/\Lambda$ (see the remark below for why such a choice of H is always possible). With this choice given we define

$$G_f(z) := \prod_{p \in H \setminus \{0\}} (\wp(z) - \wp(p))^{n_p},$$

where the exponent is defined as

$$n_p := \begin{cases} \text{ord}_p(f) & \text{if } 2p \notin \Lambda \\ \frac{1}{2} \text{ord}_p(f) & \text{if } 2p \in \Lambda \end{cases}$$

Note that this product is actually a finite product as $n_p \neq 0$ happens only at finitely many points. To complete the proof we show that $\text{ord}_p(f) = \text{ord}_p(G_f)$ for all $p \in \mathbb{C}$ which implies that the quotient f/G_f is constant by Liouville's theorem and in particular there is a constant c such that $f = c \cdot G_f \in \mathbb{C}(\wp(z), \wp'(z))$. We will first show that the orders match for all $\omega \in D \setminus \{0\}$, so let us choose such a ω . By the assumption on H , we have that exactly one of the two elements $\omega, -\omega$ is equivalent in $\mathbb{C}/\Lambda$ to an element in H . Thus there exists a $\lambda \in \Lambda$ with either $\omega + \lambda \in H$ or $-\omega + \lambda \in H$. Since f and G_f are even, we have $\text{ord}_x(f) = \text{ord}_{-x}(f)$ and $\text{ord}_x(G_f) = \text{ord}_{-x}(G_f)$ thus to compare orders, it suffices to prove that orders match when $\omega \in H$. For such ω we have

$$\text{ord}_p(G_f) = n_p \cdot \text{ord}_p(\wp(z) - \wp(p))$$

Consider the function $g_p(z) = \wp(z) - \wp(p)$. It has only one pole in D which has order 2. Therefore by Theorem 1.7 it has two zeros when counted with multiplicity. We show that when $p \notin \frac{1}{2}\Lambda$ that this function has two distinct zeros in D and that when $p \in \frac{1}{2}\Lambda$, it has a double zero at p . If $p \notin \frac{1}{2}\Lambda$, then let us choose $\lambda \in \Lambda$ such that $-p + \lambda \in D$. The assumption tells us $-p + \lambda \neq p$ and thus

$$g_p(-p + \lambda) = g_p(-p) = g_p(p) = 0.$$

Hence we have shown that g_p has two distinct zeros in D , and therefore they are both simple. Consider now the case where $p \in \frac{1}{2}\Lambda$. In this case we already know that $\text{ord}_p(g_p)$ is even and positive and at most two. The only option for this is $\text{ord}_p(g_p) = 2$. Combining these two results and using the definition, n_p we have proven

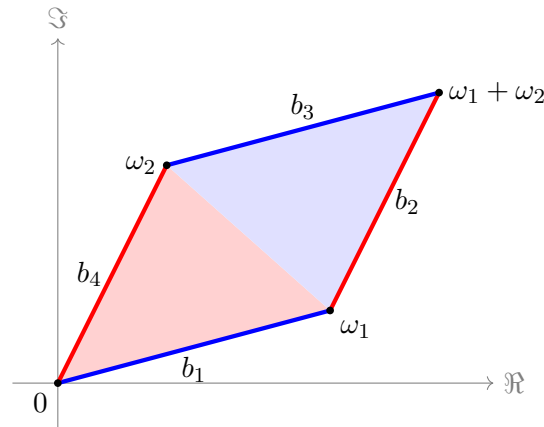
$$\text{ord}_p(f) = \text{ord}_p(G_f) \quad \text{for all } p \in H \setminus \{0\},$$

and thus also (by evenness):

$$\text{ord}_p(f) = \text{ord}_p(G_f) \quad \text{for all } p \in D \setminus \{0\}.$$

Applying theorem 1.7 this forces $\text{ord}_0(f) = \text{ord}_0(G_f)$ as well and thus the conclusion follows. $\square$

Remark 2.5. If $D = \{a + t_1\omega_1 + t_2\omega_2 \mid 0 \leq t_1, t_2 < 1\}$ for some $a \in \mathbb{C}$, then we may choose $H = \{a + t_1\omega_1 + t_2\omega_2 \mid 0 \leq t_1, t_2 < 1, t_1 + t_2 \leq 1\}$ as illustrated below



§3 Problem Set 1

Exercise 3.1. Let $\mathbb{H} = \{\tau \in \mathbb{C} \mid \Im(\tau) > 0\}$ be the Poincaré upper half-plane. For any matrix in $SL_2(\mathbb{R})$ let us set

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau := \frac{a\tau + b}{c\tau + d}$$

1. Prove that this formula defines a transitive action of $SL_2(\mathbb{R})$ on $\mathbb{H}$.
2. Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$. Prove that if $\tau' = \frac{a\tau + b}{c\tau + d}$, then the lattices $\mathbb{Z} \oplus \mathbb{Z}\tau$ and $\mathbb{Z} \oplus \mathbb{Z}\tau'$ are homothetic.

Deduce that the map

$$\begin{aligned} \mathbb{H}/SL_2(\mathbb{Z}) &\rightarrow \{\text{Lattices in } \mathbb{C}\}/\mathbb{C}^\times \\ [\tau] &\mapsto [\mathbb{Z} \oplus \mathbb{Z}\tau] \end{aligned}$$

is well-defined.

3. Prove it is a bijection.

Solution. 1. First notice that the map is well defined as the denominator has positive imaginary part and is therefore nonzero. Next, we show $SL_2(\mathbb{R}) \cdot \mathbb{H} \subseteq \mathbb{H}$. Indeed:

$$\Im\left(\frac{a\tau + b}{c\tau + d}\right) > 0 \Leftrightarrow \Im((a\tau + b)(c\bar{\tau} + d)) > 0 \Leftrightarrow \Im(\tau ad + \bar{\tau} cb) > 0,$$

The last inequality is true as can be seen by using $ad - bc = 1$ and finding:

$$\tau ad + \bar{\tau} cb = \tau ad + \bar{\tau}(ad - 1) = 2ad\Re(\tau) - \bar{\tau},$$

which clearly has positive imaginary.

We will now verify that this definition defines a group action on $\mathbb{H}$. $I_2 \cdot \tau = \tau$ is trivial from the definition. for associativity let $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\begin{pmatrix} e & f \\ g & h \end{pmatrix}$ be two matrices in $SL_2(\mathbb{R})$. Then compute :

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \left(\begin{pmatrix} e & f \\ g & h \end{pmatrix} \cdot \tau \right) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \frac{e\tau + f}{g\tau + h}$$

$$\begin{aligned}
&= \frac{\frac{e\tau+f}{g\tau+h}a + b}{\frac{e\tau+f}{g\tau+h}c + d} \\
&= \frac{\frac{\tau(ea+bg)+af+bh}{g\tau+h}}{\frac{\tau(ec+dg)+fc+dh}{g\tau+h}} \\
&= \frac{\tau(ea+bg) + af + bh}{\tau(ec+dg) + fc + dh} \\
&= \begin{pmatrix} ea+bg & fa+hb \\ ec+gd & fc+hd \end{pmatrix} \cdot \tau \\
&= \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} e & f \\ g & h \end{pmatrix} \right) \cdot \tau.
\end{aligned}$$

Finally we need to prove that this group action is transitive. First note that the matrix $\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$, maps $\tau \mapsto \tau + b$. Those for a given τ we can reach all elements of

$\mathbb{H}$ which have the same imaginary part as $\mathbb{C}$. Similarly the matrix $\begin{pmatrix} a & 0 \\ 0 & \frac{1}{a} \end{pmatrix}$, maps $\tau \rightarrow a^2\tau$, which shows that for a given τ , we can reach all elements of $\mathbb{H}$, which lie on the open ray connecting zero to τ . This clearly implies transitivity.

2. Let us write $\Lambda \cong \Lambda'$, when there exists a homothety between these two lattices. By multiplying out denominators, we have

$$\mathbb{Z} \oplus \mathbb{Z}\tau' \cong \mathbb{Z}(c\tau + d) \oplus \mathbb{Z}(a\tau + b).$$

Consider the map

$$A: \mathbb{Z} \oplus \mathbb{Z}\tau \rightarrow \mathbb{Z} \oplus \mathbb{Z}\tau, \begin{cases} 1 \mapsto c\tau + d \\ \tau \mapsto a\tau + b. \end{cases}$$

It is represented by the matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, and therefore it is invertible with integer matrix inverse. In particular, $\mathbb{Z}(c\tau + d) \oplus \mathbb{Z}(a\tau + b) = \mathbb{Z} \oplus \mathbb{Z}\tau$. In particular, the map given is well defined.

3. We need to show that the map

$$\begin{aligned}
\mathbb{H}/SL_2(\mathbb{Z}) &\rightarrow \{\text{Lattices in } \mathbb{C}\}/\mathbb{C}^\times \\
[\tau] &\mapsto [\mathbb{Z} \oplus \mathbb{Z}\tau]
\end{aligned}$$

is a bijection. It is easy to see that this map is surjective as:

$$\omega_1\mathbb{Z} \oplus \omega_2\mathbb{Z} \cong \mathbb{Z} \oplus \frac{\omega_2}{\omega_1}\mathbb{Z} \cong \mathbb{Z} \oplus \pm \frac{\omega_2}{\omega_1}\mathbb{Z}$$

Where we see that one of the two elements $\pm \frac{\omega_1}{\omega_2}$ will have positive imaginary part. Suppose now we have two homothetic lattices:

$$\mathbb{Z} \oplus \mathbb{Z}\tau \cong \mathbb{Z} \oplus \mathbb{Z}\tau'.$$

then we may choose a nonzero complex number γ with

$$\mathbb{Z} \oplus \mathbb{Z}\tau = \mathbb{Z}\gamma \oplus \mathbb{Z}\tau'\gamma.$$

In particular, we have

$$\begin{aligned}\gamma &= a + b\tau \\ \gamma\tau' &= c + d\tau,\end{aligned}$$

for integers a, b, c, d with $ac - bd = \pm 1$. In fact $ac - bd = 1$ as the change of basis matrix from $\{1, \tau\}$ to $\{\gamma, \tau'\gamma\}$ preserves orientation. dividing these two we have:

$$\tau' = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau.$$

□

Exercise 3.2. Let $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ be a lattice and $\mathbb{C}/\Lambda$ the associated elliptic curve.

- (1) Prove that $\mathbb{C}/\Lambda$ is isomorphic to

$$\mathbb{C}/(\mathbb{Z} \oplus \mathbb{Z}\tau), \quad \text{with } \tau = \omega_1/\omega_2.$$

- (2) Prove that the endomorphisms of $\mathbb{C}/\Lambda$ form an integral extension of $\mathbb{Z}$, that is to say a ring containing $\mathbb{Z}$ whose elements are all roots of monic polynomials with integer coefficients.
- (3) Prove that if $\text{End}(\mathbb{C}/\Lambda)$ is not reduced to $\mathbb{Z}$, then

$$K = \mathbb{Q}(\tau)$$

is an imaginary quadratic field and there is an isomorphism

$$\text{End}(\mathbb{C}/\Lambda) \otimes_{\mathbb{Z}} \mathbb{Q} \cong K.$$

One then says that the elliptic curve $\mathbb{C}/\Lambda$ has **complex multiplication**.

- (4) Give examples of elliptic curves with complex multiplication.

Proof. (1) Consider the isomorphism of $\mathbb{C}$

$$\tau: \mathbb{C} \rightarrow \mathbb{C} \quad x \mapsto \frac{x}{\omega_1},$$

this map, maps $\tau(\mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2) = \mathbb{Z} \oplus \mathbb{Z}\tau$. In particular we have a group isomorphism

$$\mathbb{C}/(\mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2) \rightarrow \mathbb{C}/(\mathbb{Z} \oplus \mathbb{Z}\tau),$$

which is a biholomorphism as it comes a biholomorphism of $\mathbb{C}$ which is the universal cover of both spaces.

- (2)+(3) by (1) it suffices to consider lattices of the form $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$. We know from the lecture that

$$\text{End}(\mathbb{C}/\Lambda) \cong \{\alpha \in \mathbb{C} \mid \alpha(\Lambda) \subseteq \Lambda\}$$

Interpreting α as a map on the $\mathbb{Z}$ module Λ , Nakayamas lemma ensures that α satisfies a monic equation over the integers of degree 2. Hence $\text{End}(\mathbb{C}/\Lambda)$ defines an integral extension of $\mathbb{Z}$. Infact, there exists integers a, b, c, d , such that

$$\alpha \cdot 1 = a + b\tau, \quad \alpha \cdot \tau = c + d\tau.$$

The first equation says that $\alpha \in \mathbb{Z}[\tau]$. Combining both equations, we see (assuming $\alpha \neq 0$):

$$\tau = \frac{\tau\alpha}{\alpha} = \frac{c + d\tau}{a + b\tau}.$$

Multiplying out denominators, we see that $\mathbb{Q}(\tau)$ is a extension of degree 2 over $\mathbb{Q}$. and $\mathbb{Z} \subseteq \text{End}(\mathbb{C}/\Lambda) \subseteq \mathbb{Z}[\tau]$. Tensoring this relation with $\mathbb{Q}$, we obtain:

$$\mathbb{Q} \subseteq \text{End}(\mathbb{C}/\Lambda) \otimes_{\mathbb{Z}} \mathbb{Q} \subseteq K.$$

Since the extension $K/\mathbb{Q}$ has degree 2, $\text{End}(\mathbb{C}/\Lambda) \otimes_{\mathbb{Z}} \mathbb{Q}$ is either K or $\mathbb{Q}$. If $\text{End}(\mathbb{C}/\Lambda)$ is not $\mathbb{Z}$, then the second case is impossible since localization is exact.

- (4) Consider the Gaussian integers: $\Lambda = \mathbb{Z} \oplus i\mathbb{Z}$. They have multiplication by i , and in fact $\text{End}(\mathbb{C}/\Lambda) \cong \mathbb{Z}[i]$.

Remark 3.3. Notice that the multiplication $\alpha \cdot \Lambda$ gives a fractional ideal of $\mathbb{Z}[\tau]$. The condition $\alpha \cdot \Lambda \subseteq \Lambda$, means that this fractional ideal is actually an ideal. Hence there is some arithmetic hidden in what is happening. In fact, there is a connection to the picard group, which we will explore later.

□

Exercise 3.4. Let $\Lambda \subset \mathbb{C}$ be a lattice with invariants g_2 and g_3 , and let $z_1, z_2 \in \mathbb{C} \setminus \Lambda$ be complex numbers such that $\wp(z_1) \neq \wp(z_2)$. Let

$$y = ax + b$$

be the equation of the line passing through the points $(\wp(z_1), \wp'(z_1))$ and $(\wp(z_2), \wp'(z_2))$.

- (1) Prove that the function $\wp'(z) - a\wp(z) - b$ has three zeros counted with multiplicity.
 (2) Deduce that if $2z_1 + z_2$ or $z_1 + 2z_2$ do not belong to the lattice, then this function has a zero at a point

$$z_3 = -z_1 - z_2 \pmod{\Lambda}.$$

- (3) Prove the equality

$$4(x - \wp(z_1))(x - \wp(z_2))(x - \wp(z_3)) = 4x^3 - g_2x - g_3 - (ax + b)^2.$$

- (4) Deduce the equality

$$\wp(z_1) + \wp(z_2) + \wp(z_3) = \frac{a^2}{4}.$$

- (5) Prove the equality for all $z_1, z_2 \in \mathbb{C}$:

$$\wp(z_1 + z_2) = -\wp(z_1) - \wp(z_2) + \frac{1}{4} \left(\frac{\wp'(z_1) - \wp'(z_2)}{\wp(z_1) - \wp(z_2)} \right)^2.$$

- (6) By letting z_2 approach z_1 , deduce the duplication formula:

$$\wp(2z) = -2\wp(z) + \frac{1}{4} \left(\frac{\wp''(z)}{\wp'(z)} \right)^2.$$

- (7) Let (x_1, y_1) and (x_2, y_2) be two distinct points on the affine algebraic curve

$$C : y^2 = 4x^3 - g_2x - g_3.$$

Prove that there exists a point (x_3, y_3) in the intersection between C and the line passing through (x_1, y_1) and (x_2, y_2) with coordinate

$$x_3 = -x_1 - x_2 + \frac{1}{4} \left(\frac{y_1 - y_2}{x_1 - x_2} \right)^2$$

and compare with the preceding.

Exercise 3.5. Let $\Lambda \subset \mathbb{C}$ be a lattice. Let δ denote the minimal distance between two distinct points of Λ and V the volume of a fundamental domain.

- (1) Let A be an annulus of inner radius r and thickness $\delta/2$. Prove that $A \cap \Lambda$ contains at most $4\pi r/\delta$ points.
- (2) Prove the estimate

$$|\{\omega \in \Lambda \mid |\omega| \leq R\}| = \frac{\pi R^2}{V} + O(R) \quad \text{as } R \rightarrow +\infty.$$

- (3) Deduce that there exists a constant c depending only on Λ such that

$$|\{\omega \in \Lambda \mid R \leq |\omega| < R+1\}| \leq cR \quad \text{for all } R > 0.$$

Exercise 3.6. Let ω_1, ω_2 be two complex numbers with $\Im(\omega_2/\omega_1) > 0$ and $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ the lattice they generate. Consider the series

$$\zeta_\Lambda(z) = \frac{1}{z} + \sum_{\omega \in \Lambda \setminus \{0\}} \left(\frac{1}{z - \omega} + \frac{1}{\omega} + \frac{z}{\omega^2} \right).$$

- (1) Prove that $\zeta_\Lambda(z)$ converges absolutely for all $z \in \mathbb{C} \setminus \Lambda$ and defines a meromorphic function on $\mathbb{C}$ having simple poles of residue 1 at the points of Λ .
- (2) Is ζ_Λ an elliptic function?
- (3) Prove the equality

$$\frac{d}{dz} \zeta_\Lambda = -\wp_\Lambda,$$

where $\wp_\Lambda$ denotes the Weierstrass function associated to Λ .

- (4) Let $\omega_3 = \omega_1 + \omega_2$. Prove that for $j = 1, 2, 3$, there exist $\eta_j \in \mathbb{C}$ such that

$$\zeta_\Lambda(z + \omega_j) = \zeta_\Lambda(z) + \eta_j$$

for all $z \in \mathbb{C} \setminus \Lambda$, and then the equality

$$\eta_j = 2\zeta_\Lambda(\omega_j/2).$$

- (5) Using Cauchy's residue theorem, prove the equality

$$\eta_1\omega_2 - \eta_2\omega_1 = 2i\pi.$$

§4 Lecture 3

Last week we introduced the Weierstrass $\wp$ function of a lattice:

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda \setminus \{0\}} \left(\frac{1}{z - \omega^2} - \frac{1}{\omega^2} \right).$$

We have seen that all elliptic functions are generated by it and its derivative and so this function determines a lot of information for the lattice. Let us investigate this function more closely. We would like to calculate its Laurent series at 0, and for this notice we have

$$\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} = \frac{1}{\omega^2} \left(\frac{1}{(\frac{z}{\omega} - 1)^2} - 1 \right),$$

Notice that for $|t| < 1$, we have

$$\frac{1}{(1 - t)^2} = \sum_{n=1}^{\infty} n t^{n-1}.$$

Thus, for $|z| < |\omega|$ we have

$$\frac{1}{\omega^2} \left(\frac{1}{(\frac{z}{\omega} - 1)^2} - 1 \right) = \frac{1}{\omega^2} \left(\sum_{n=0}^{\infty} \left((n+1) \left(\frac{z}{\omega} \right)^n \right) - 1 \right) = \frac{1}{\omega^2} \sum_{n=1}^{\infty} (n+1) \left(\frac{z}{\omega} \right)^n$$

Which can be rewritten in a power series in z as:

$$\sum_{n=1}^{\infty} \left(\frac{n+1}{\omega^{n+2}} \right) z^n.$$

Plugging this into the Weierstrass function, we get

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda \setminus \{0\}} \left(\sum_{n=1}^{\infty} \frac{n+1}{\omega^{n+2}} z^n \right) = \frac{1}{z^2} + \sum_{n=1}^{\infty} z^n \sum_{\omega \in \Lambda \setminus \{0\}} \frac{n+1}{\omega^{n+2}}$$

where we may sum using absolute convergence. Splitting the above sum into two parts we receive:

$$\wp(z) = \frac{1}{z^2} + \sum_{n=1}^{\infty} z^{2n} (2n+1) G_{2(n+1)} + \sum_{n=1}^{\infty} z^{2n-1} 2n \sum_{\omega \in \Lambda \setminus \{0\}} \frac{1}{\omega^{2n+1}} = \frac{1}{z^2} + \sum_{n=1}^{\infty} z^{2n} (2n+1) G_{2(n+1)}$$

where in the last line we used that

$$\sum_{\omega \in \Lambda \setminus \{0\}} \frac{1}{\omega^{2n+1}} = 0$$

as for any $\omega \in \Lambda \setminus \{0\}$, its contribution to the above sum is canceled out by the contribution of $-\omega$. Using our new Laurent expansion:

$$\wp(z) = \frac{1}{z^2} + \sum_{n=1}^{\infty} z^{2n} (2n+1) G_{2(n+1)},$$

which is valid near zero, we can deduce the following:

Proposition 4.1

for all $z \in \mathbb{C} \setminus \Lambda$, we have

$$\wp'(z)^2 = 4\wp(z)^3 - 60G_4\wp(z) - 140G_6.$$

Remark 4.2. We denote the numbers $60G_4$ and $140G_6$, appearing in the above equations, as

$$\begin{aligned} g_2 &:= 60G_4 \\ g_3 &:= 140G_6. \end{aligned}$$

They are called invariants of Λ .

Proof. Let $F(z) := \wp'(z) - 4\wp(z)^3 + 60G_4\wp(z) + 140G_6$. We will show that F is holomorphic, which would imply, since F is elliptic, that F is constant. For each of the terms appearing in F we calculate the Laurent series. First, we have:

$$\begin{aligned} \wp'(z) &= -\frac{2}{z^3} + \sum_{n=1}^{\infty} z^{2n-1} 2n \cdot (2n+1) \cdot G_{2n+2} \\ \Rightarrow \wp'(z)^2 &= \frac{4}{z^6} - \frac{24}{z^2} G_4 - 80G_6 + o(z). \end{aligned}$$

Similarly we may calculate

$$\wp(z)^3 = \frac{1}{z^6} + \frac{9}{z^2} G_4 + 15G_6 + o(z)$$

Using these three expansions, we conclude that the Laurent series expansion of F is given by:

$$\begin{aligned} F(z) &= \left(\frac{4}{z^6} - \frac{24}{z^2} G_4 - 80G_6 + o(z) \right) - 4 \left(\frac{1}{z^6} + \frac{9}{z^2} G_4 + 15G_6 + o(z) \right) + 60 \left(\frac{1}{z^2} + o(z) \right) + 140G_6 \\ &= o(z) \end{aligned}$$

Thus, the conclusion. □

Using the above theorem is that it allows us to construct a map

$$\mathbb{C} \setminus \Lambda \rightarrow \{(x, y) \in \mathbb{C}^2 \mid y^2 = 4x^3 - g_2x - g_3\}, \quad z \mapsto (\wp(z), \wp'(z)).$$

In fact we can projectivize to get a holomorphic map

$$\mathbb{P}\mathbb{C}^2 \rightarrow E := \{[x : y : z] \in \mathbb{C}^2 \mid y^2 z = 4x^3 - g_2 x z^3 - g_3 z^3\}, \quad z \mapsto \begin{cases} [\wp(z) : \wp'(z) : 1] & \text{if } z \in \mathbb{C} \setminus \Lambda \\ [0 : 1 : 0] & \text{if } z \in \Lambda. \end{cases}$$

Proposition 4.3

The polynomial $f(x) = 4x^3 - g_2x - g_3$ has three distinct roots and in particular $\Delta = g_2^3 - 27g_3^2 \neq 0$.

Proof. let us choose a $\mathbb{Z}$ basis $\Lambda = \omega_1\mathbb{Z} \oplus \omega_2\mathbb{Z}$ and define $\omega_3 = \omega_1 + \omega_2$. Since $\wp'$ is an odd function we obtain:

$$\wp'\left(\frac{\omega_i}{2}\right) = -\wp'\left(-\frac{\omega_i}{2}\right) = -\wp'\left(\frac{\omega_i}{2}\right),$$

which implies that $\wp'$ vanishes at all three ω_i . In particular, by the map constructed above, the elements $\wp(\omega_i/2)$ are all roots of the polynomial f . It suffice to show that they are all distinct. Consider the function

$$g(z) = \wp(z) - \wp\left(\frac{\omega_i}{2}\right),$$

It is elliptic and has a pole of order two. Thus it can have only two zeros with multiplicity. However, $\omega_i/2$ is already a root of multiplicity 2 which implies that the others are not and we obtain $\wp(\omega_i/2) \neq \wp(\omega_j/2)$ for $j \neq i$. $\square$

Using this we can make a stronger statement about our map to E which we constructed above.

Proposition 4.4

The map

$$\varphi: \mathbb{C}/\Lambda \rightarrow E, z \mapsto \begin{cases} [\wp(z) : \wp'(z) : 1] & \text{if } z \in \mathbb{C} \setminus \Lambda \\ [0 : 1 : 0] & \text{if } z \in \Lambda. \end{cases}$$

is a biholomorphism

Proof. Because φ is a holomorphic map, its open an open map, and because $\mathbb{C}/\Lambda$ is compact, its image is also closed. If we show that φ is bijective, it will automatically cause φ to be a homeomorphism since it is a continuous bijection between compact sets. Using the inverse mapping theorem, we see that in this case φ is actually a biholomorphism since, locally in charts, φ is represented with $\varphi(0) = 0$ and $\varphi = z^m + a_1 z^{m+1} + \dots$, $m \geq 1$. If $m > 1$, this would fail to be injective locally at zero and thus $m = 1$ and $\varphi'(0) \neq 0$. Hence, we only need to show that φ is a bijection.

Let us show first, that φ is an injective map. Lets $z_1, z_2 \in \mathbb{C}/\Lambda$ such that $\varphi(z_1) = \varphi(z_2)$ and $\varphi'(z_1) = \varphi'(z_2)$. We need to show that $z_1 \equiv z_2 \pmod{\Lambda}$. This is clear if

$$\varphi(z_1) = [0 : 1 : 0] = \varphi(z_2),$$

thus, let us assume we are in the affine $z \neq 0$. Then we have

$$\begin{aligned} \wp(z_1) &= \wp(z_2) \\ \wp'(z_1) &= \wp'(z_2) \end{aligned}$$

Now, consider the function

$$g(z) = \wp(z) - \wp(z_1),$$

If $2z_1 \notin \Lambda$, it vanishes at $z_1, -z_1$ and z_2 . As it only has one pole, with order, two in $\mathbb{C}/\Lambda$, we receive $z_2 = \pm z_1$. Assuming $z_2 = -z_1$, we then arrive at a contradiction for

$$\wp'(z_2) = \wp'(-z_1) = -\wp'(z_1) \Rightarrow \wp'(z_1) = 0$$

which means that z_1 is a double root and hence the only one, which implies $2z_1 \in \Lambda$. If $2z_1 \in \Lambda$, then we may verify that z_1 is a double root of g just as above, which implies $z_2 = z_1$.

Now let us show surjectivity. Once again, it suffices to show surjectivity on the affine

$$V(y^2 - 4x^3 - g_2x - g_3) \subseteq E,$$

as the only point of $V(y^2z - 4x^3 - g_2xz^2 - g_3z^3)$ with $z = 0$ is $[0 : 1 : 0]$ which is in the image of φ . Hence suppose we have (x, y) with

$$y^2 - 4x^3 - g_2x - g_3 = 0.$$

The function

$$g(z) = \wp(z) - x,$$

is nonconstant elliptic and thus, has a zero. Therefore $x = \wp(a)$ for some $a \in \mathbb{C}$. as $\wp'(a)$ solves the equation

$$t^2 - 4x^3 - g_2x - g_3 = 0,$$

and so does y , we receive $y = \pm\wp'(a)$. if $y = \wp'(a)$ we are done and otherwise, $y = \wp'(-a)$ and we also have $x = \wp(-a)$. $\square$

Remark 4.5. We have already shown that the curve $V(y^2z - 4x^3 + g_2xz^2 + g_3z^3) \subseteq \mathbb{P}\mathbb{C}^2$ is smooth

Now we have started from lattices and arrived at a cubic equation (elliptic curve). The next theorem says that we can go the otherway around as well:

Theorem 4.6 (Uniformization theorem)

let $a, b \in \mathbb{C}$ such that $a^3 - 27b^2 \neq 0$. Then there exists a lattice $\Lambda \subseteq \mathbb{C}$ with $a = g_2(\Lambda)$, and $b = g_3(\Lambda)$. In other words, any Riemann surface $E \subseteq \mathbb{P}\mathbb{C}^2$ of the form

$$E : y^2z = 4x^3 - axz^2 - bz^3$$

is biholomorphic to a Riemann surface $\mathbb{C}/\Lambda$. The (covering) map $\Phi: \mathbb{C} \rightarrow E$, which makes

$$\begin{array}{ccc} & \mathbb{C} & \\ \pi \swarrow & & \searrow \Phi \\ \mathbb{C}/\Lambda & \xrightarrow{\cong} & E \end{array}$$

commute, is called uniformization map.

Remark 4.7. We will use as a black box that E defined as above is a torus and has fundamental group $\pi_1(E) \cong \mathbb{Z} \times \mathbb{Z}$. This follows for example, from the genus formula.

(*sketch*). We already have a map in one direction: given a lattice, we have constructed E in proposition 4.4. Conversely, suppose that we have

$$E : y^2z = 4x^3 - axz^2 - bz^3, \quad a^3 - 27b^2 \neq 0$$

Then E is a smooth Riemann surface. Now consider the one-form

$$\alpha = \frac{dx}{y}.$$

on E . We claim, it is holomorphic on E . this is obvious for $y \neq 0$. Thus, we need to consider the set

$$E \cap \{y = 0\} = \{[x : 0 : z] \mid 4x^3 - axz^2 - bz^3 = 0\}$$

We see that if $z = 0$ we obtain $x = 0$, which is not possible, thus we can restrict to the affine $\{z = 1\}$, as being holomorphic is a local property. In these coordinates, we are looking at the curve

$$E' = \{(x, y) \mid y^2 = 4x^3 - ax - b\},$$

Let $f(p) = 4x^3 - ax - b$. The assumption states that f does not have double zeros, and hence $f'(a) \neq 0$ for all $a \in E'$. differentiating the governing equation of E we obtain

$$d(y^2) = 2y \cdot dy = f'(x) \cdot dx.$$

Therefore,

$$\frac{dx}{y} = \frac{2dy}{f'(a)},$$

outside of $y = 0$ and as the right side is defined at $y = 0$ as well, α extends to a global holomorphic differential. denote by $O = [0 : 1 : 0]$ and let γ_1, γ_2 be generators of the fundamental group of E with base point O , i.e

$$\pi_1(E, O) = \gamma_1 \mathbb{Z} \oplus \gamma_2 \mathbb{Z}.$$

Define the complex numbers ω_1, ω_2 by

$$\omega_i = \int_{\gamma_i} \frac{dx}{y}.$$

We claim that these ω_i define a lattice. Suppose for sake of contradiction that the ω_i lie on a real line in $\mathbb{C}$. If we multiply $\alpha = \frac{dx}{y}$ by a constant, we may assume that they are all real. Then the map

$$h : P \mapsto \text{Im}\left(\int_O^P \alpha\right),$$

is independent of the choice of path from O to P , since changing the path results in a change in period in $\mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2 \subseteq \mathbb{R}$. This function is therefore a differentiable function on a compact Riemann surface and hence its image is closed. On the other hand, it is the imaginary part of a holomorphic function, and thus its image is open as well. Thus, the function is constant. differentiating, we obtain

$$dh = \text{Im}(\alpha) = 0.$$

But since α is holomorphic, this forces $\alpha = 0$ which is clearly not the case. Hence, we may define the lattice

$$\Lambda := \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2 \subseteq \mathbb{C}.$$

Finally, we can define the map

$$A : E \rightarrow \mathbb{C}/\Lambda, P \mapsto \left[\int_O^P \frac{dx}{y} \right] \mod \Lambda.$$

The value of this is independent of choice of path between O and P for two different paths, the difference of the respective integrals is a period in Λ by construction. Now

that we have constructed $\mathbb{C}/\Lambda$, the Weierstraß embedding gives us another elliptic curve E' such that $E' \cong \mathbb{C}/\Lambda$. We claim $E = E'$ which would finish the claim.

Note now the the function

$$z: E \rightarrow \mathbb{C}, p \mapsto \int_O^p \frac{dx}{y},$$

is defined locally near O . This z serves as a local uniformizer at O : we have $z(O) = 0$ and $dz = \frac{x}{y} \Rightarrow y(z) = \frac{dx}{dt} = x'(t)$. We claim x has a pole of order 2 at O and is an even function.

$$x(z) = \frac{1}{z^2} + c_2 z^2 + c_4 z^6 + \mathcal{O}(z^6),$$

and

$$y(z) = x'(z) = \frac{1}{z^3} + 2c_2 z + 4c_4 z^3 + \mathcal{O}(z^5)$$

and so

$$y(z)^2 = 4x(z)^3 - ax(z) - b,$$

and hence

$$c_2 = a/20, \text{ and } c_4 = \frac{b}{28},$$

and also

$$\wp_\Lambda(z) = \frac{1}{z^2} + \frac{g_2}{20z^2} + \frac{g_3}{28}z^4 + \mathcal{O}(z^6).$$

therefore $x(z) = p(z)$ and the claim about a and b . □

Definition 4.8. We call Riemann surfaces of the form $\mathbb{C}/\Lambda$, complex elliptic curves.

§5 Lecture 4

Consider for a moment two lattices $\Lambda_1, \Lambda_2 \in \mathbb{C}$ and let $\alpha \in \mathbb{C}$, such that $\alpha(\Lambda_1) \subseteq \Lambda_2$. The homomorphism theorem of groups, tells us the existence of a unique group homomorphism $f_\alpha: \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2$, such that the following diagram commutes.

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{\alpha} & \mathbb{C} \\ \downarrow \pi & & \downarrow \pi \\ \mathbb{C}/\Lambda_1 & \xrightarrow{f_\alpha} & \mathbb{C}/\Lambda_2 \end{array}$$

In fact, it is not hard to see that this f_α is a holomorphic map, since it locally looks like scaling by α , since π are covering maps. The following theorem tells us that up to translation, all holomorphic maps between these two complex elliptic curves are obtained in this way.

Theorem 5.1

There is a bijection

$$\begin{aligned} \{\alpha \in \mathbb{C} \mid \alpha(\Lambda_1) \subseteq \Lambda_2\} &\rightarrow \{\text{holomorphic maps } f: \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2, f(0) = 0\} \\ \alpha &\mapsto f_\alpha. \end{aligned}$$

Proof. The map is well defined by the comments above and is also holomorphic: for $p \in \mathbb{C}/\Lambda$, choose a neighborhood U of p small enough such that $\pi^{-1}(U) \cong U \times D$ for some discrete set U . Let $g: U \rightarrow U_\alpha$ be a local inverse to g , which is by the construction of the differentiable structure on $\mathbb{C}/\Lambda$, a biholomorphism. Then $f_\alpha|_U = \pi \circ \alpha \circ g$ is a composition of holomorphic maps.

Now let us show that the correspondence defined in the theorem actually defines a bijection.

Injectivity: Suppose $\alpha, \beta \in \mathbb{C}$ with $\alpha(\Lambda_1) \subseteq \Lambda_2$, $\beta(\Lambda_1) \subseteq \Lambda_2$ and $f_\alpha = f_\beta$. This means, that for all $z \in \mathbb{C}$, we have

$$\alpha z \equiv \beta z \pmod{\Lambda_2}.$$

In particular, we have

$$(\alpha - \beta)\mathbb{C} \subseteq \Lambda_2.$$

Since Λ_2 is discrete, this means $\alpha - \beta$ is a constant map. Furthermore, $(\alpha - \beta)(0) = 0$. Thus, $\alpha = \beta$.

Surjectivity: Recall The lifting property from topology: Suppose

$$p: (X', x'_0) \rightarrow (X, x_0),$$

is a covering space of pointed topological spaces, and consider a continuous map of pointed spaces

$$g: (Y, y_0) \rightarrow (X, x_0)$$

Then there exists a lifting;

$$\begin{array}{ccc} & & (X', x'_0) \\ & \nearrow \tilde{g} & \downarrow p \\ (Y, y_0) & \xrightarrow{g} & (X, x_0) \end{array}$$

if and only if $g_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(X', x'_0))$. In this case the lifting is unique. Let us apply this to our setting. Assume we have a holomorphic map

$$f: \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2,$$

With $f(0) = 0$. Then, the lifting property tells us of a unique map $\tilde{f}$ such that the following diagram commutes:

$$\begin{array}{ccc} (\mathbb{C}, 0) & \xrightarrow{\tilde{f}} & (\mathbb{C}, 0) \\ \downarrow \pi & & \downarrow \pi \\ (\mathbb{C}/\Lambda_1, [0]) & \xrightarrow{f} & (\mathbb{C}/\Lambda_2, [0]). \end{array}$$

First note, that $\tilde{f}$ is holomorphic, since it locally looks like f . Furthermore, $\tilde{f}$ satisfies

$$\tilde{f}(z + \omega) \equiv \tilde{f}(z) \pmod{\Lambda_2},$$

for all $\omega \in \Lambda_1$. In particular, the map $g_\omega(z) := \tilde{f}(z + \omega) - \tilde{f}(z)$ maps to the discrete set Λ_2 , and is therefore constant say $g_\omega(z) = c(\omega)$. If we differentiate, we find

$$\tilde{f}'(z + \omega) = \tilde{f}'(z).$$

Thus, $\tilde{f}'(z)$ is an elliptic holomorphic function and therefore constant, and $\tilde{f}(z) = \alpha z + b$. Using $\tilde{f}(0) = 0$, we obtain the result.

□

Remark 5.2. The above theorem shows, in particular, that all holomorphic maps

$$f: \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_1, \text{ with } f(0) = 0,$$

are group homomorphisms.

Let us apply this theorem to investigate the endomorphism ring $\text{End}(\mathbb{C}/\Lambda)$.

Proposition 5.3

Let $\mathbb{C}/\Lambda$ be a complex elliptic curve, then we have

$$\text{End}(\mathbb{C}/\Lambda) \cong \begin{cases} \mathbb{Z} \\ \text{or} \\ \text{an order in an imaginary quadratic field } A \subseteq k, \end{cases}$$

Where $\text{End}(\mathbb{C}/\Lambda)$ is defined as the set of holomorphic maps $f: \mathbb{C}/\Lambda \rightarrow \mathbb{C}/\Lambda$, which map zero to zero. In the second case we say that $\mathbb{C}/\Lambda$ has complex multiplication.

Remark 5.4. Recall that $\mathcal{O} \subseteq k$ is called an **order** if

- $\mathcal{O}$ is a subring of k
- $\mathcal{O}$ is a $\mathbb{Z}$ -lattice of k , meaning $k \cong \mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Q}$.

Note that this means that $\mathcal{O}$ is a free $\mathbb{Z}$ -module of rank $\dim_{\mathbb{Q}}(k)$.

Proof of Proposition 5.3. By Theorem 5.1 we have

$$\text{End}(\mathbb{C}/\Lambda) \cong \{\alpha \in \mathbb{C} \mid \alpha\Lambda \subseteq \Lambda\}.$$

Thus we immediately have an inclusion

$$\mathbb{Z} \hookrightarrow \text{End}(\mathbb{C}/\Lambda).$$

Choose now a $\mathbb{Z}$ -basis: $\Lambda = \omega_1\mathbb{Z} \oplus \omega_2\mathbb{Z}$. Then setting $\tau = \frac{\omega_1}{\omega_2}$, we have an isomorphism

$$\frac{\mathbb{C}}{\omega_1\mathbb{Z} \oplus \omega_2\mathbb{Z}} \cong \frac{\mathbb{C}}{\tau\mathbb{Z} \oplus \mathbb{Z}},$$

which is induced by the multiplication by $\frac{1}{\omega_2}$. Hence let us assume wlog, that Λ is of the form $\mathbb{Z} \oplus \tau\mathbb{Z}$. Then the condition $\alpha\Lambda \subseteq \Lambda$, is equivalent to asserting the existence of integers a, b, c, d with

$$\begin{aligned} \alpha \cdot 1 &= a + b\tau \\ \alpha \cdot \tau &= c + d\tau. \end{aligned}$$

Since $\tau \notin \mathbb{Z}$, we have $b \neq 0$ and thus we can rearrange the first equation to find

$$\tau = \frac{\alpha - a}{b}.$$

Plugging this equation into the second, we have

$$\alpha \cdot \frac{\alpha - a}{b} = c + d\tau.$$

clearing denominators:

$$\alpha^2 - (a + d)\alpha + (ad - bc) = 0.$$

In particular $\text{End}(\mathbb{C}/\Lambda)$ is an integral extension of $\mathbb{Z}$. Now suppose $\alpha \neq 0$, then we have

$$\tau = \frac{\alpha\tau}{\alpha} = \frac{c + d\tau}{a + b\tau} \Rightarrow b\tau^2 + \tau(a - d) + c = 0.$$

Since $\tau \notin \mathbb{R}$, $b \neq 0$, and thus $k := \mathbb{Q}(\tau)$ is a quadratic imaginary field. We have shown even more:

$$\text{End}(\mathbb{C}/\Lambda) \subseteq \mathbb{Z}[\tau] \subset k.$$

Tensoring these inclusions with $\mathbb{Q}$ over $\mathbb{Z}$ we find the desired result. $\square$

Example 5.5

Consider $\mathbb{C}/(\mathbb{Z} \oplus \sqrt{-1} \cdot \mathbb{Z})$. It has complex multiplication by $\sqrt{-1}$ and

$$\text{End}(\mathbb{C}/(\mathbb{Z} \oplus \sqrt{-1} \cdot \mathbb{Z})) = \mathbb{Z}[\sqrt{-1}].$$

As we have seen In the previous lecture, elliptic curves are given up to isomorphism by

$$\mathbb{C}/\Lambda \cong \{y^2z = 4x^3 - g_2xz^2 - g_3z^3\} \subseteq \mathbb{P}\mathbb{C}^2.$$

The RHS can be interpreted for any field, for example, the rational numbers or local fields $\mathbb{Q}_{(p)}$. Inspired from the theory of complex elliptic curves, we will now try to produce a more general theory.

§5.1 Smooth projective curves

We will recall some basic theory from algebraic geometry. Most results are basic, but what is new is the treatment of non-algebraically closed fields. The moral of the theory is that we do geometry in the algebraic closure and then keep track of the "rational" points by looking of galois invariants. From now on, we will work on a field k which we assume to be perfect, which means $\text{char}(k) = 0$ or $\text{char}(k) = p$ and The Frobenius map is an automorphism. For such a field k , fix an algebraic closure $\bar{k} \supseteq k$ and let G_k denote the Galois group $\text{Gal}(\bar{k}/k)$. Recall that for a homogeneous ideal $I \subseteq k[x_0, \dots, x_n]$ we have a well defined projective vanishing $V_{\mathbb{P}^n}(I) \subseteq \mathbb{P}^n$. Conversely for a subset S of $\mathbb{P}^n$, we can define its homogeneous ideal $I(S) \subseteq k[x_0, \dots, x_n]$.

Definition 5.6. For an algebraic set $V \subseteq \mathbb{P}^n$, we say that V is defined over k (written V/k), if $I(V)$ has generators in the subring $k[x_0, \dots, x_n] \subseteq \bar{k}[x_0, \dots, x_n]$. In this case, we define

$$V(k) = V \cap \mathbb{P}^n(k),$$

where

$$\mathbb{P}^n(k) = \{[x_0 : \dots : x_n] \mid x_i \in k \forall i\}$$

Definition 5.7. Suppose V is defined over k , then the points $V(k)$ are called **k rational points** of V .

Remark 5.8. Note that G_k acts on $\mathbb{P}^n$, via the formula

$$\sigma([x_0 : \dots : x_n]) := [\sigma(x_0) : \dots : \sigma(x_n)].$$

In fact, we have $\mathbb{P}^n(k) = (\mathbb{P}^n)^{G_k}$, where $(\mathbb{P}^n)^{G_k}$ denotes the fixed points under the above

group action. Indeed,

$$\mathbb{P}^n(k) \subseteq (\mathbb{P}^n)^{G_k}$$

is easy. For the reverse, scale so that some coordinate is one, and note that since k is perfect, every element not in k , is not fixed by some element in the Galois group. In particular, we have

$$V(k) = \{x \in V \mid \sigma(v) = v, \forall \sigma \in G_k\}.$$

We call V a projective variety if $I(V)$ is a prime ideal and if this ideal has generators in $k[x_0, \dots, x_n]$, we say it is a projective variety over k .

Let $\mathbb{A}_k^n \hookrightarrow \mathbb{P}_k^n$ be a copy of affine space such that $V \cap \mathbb{A}_k^n \neq \emptyset$. Then the function field of V is

$$\bar{k}(V) = \text{Frac} \left(\frac{k[t_1, \dots, t_n]}{I(V \cap \mathbb{A}_k^n)} \right).$$

Remark 5.9. Recall, for example,

$$\mathbb{C}(\mathbb{C}/\Lambda) = \text{Frac} \left(\frac{\mathbb{C}[x, y]}{(y^2 - 4x^3 + g_2x - g_3)} \right)$$

and $\dim(\mathbb{C}(\mathbb{C}/\Lambda)) = 1$, where the dimension here is the degree of transcendence over $\mathbb{C}$.

In general, we define the dimension of V as the degree of transcendence of its function field

. Note that the Galois group G_k acts on $\bar{k}[x_1, \dots, x_n]$, coefficient wise. Suppose now that V is defined over k and consider the composition, then for any $\sigma \in G_k$, we have $\sigma(I(V)) = I(V)$. Thus, the elements of the galois group define a well defined action on $\bar{k}(V)$. This allows us to define

$$k(V) = \bar{k}(V)^{G_k}.$$

Furthermore, for any $f \in \bar{k}(V)$, any $\sigma \in G_k$ and a point $P \in V$, for which f is regular, we have

$$\sigma(f(P)) = (\sigma(f))(\sigma(P)).$$

Remark 5.10. From elementary algebraic geometry, we could have also defined the function field as quotients of homogeneous polynomials

$$\frac{F(x_0, \dots, x_n)}{G(x_0, \dots, x_n)}, \text{ with } \deg G = \deg F, G \neq 0.$$

with

$$\frac{F}{G} = \frac{F'}{G'} \Leftrightarrow FG' - F'G \in I(V).$$

Recall, furthermore, that for a point $p \in V$, we have a local ring

$$\mathcal{O}_{V,p} := \left\{ \frac{f}{g} \in \bar{k}(V) \mid g(P) \neq 0 \right\}$$

We can also obtain the ring (up to iso) localizing the affine coordinate ring $\bar{k}[V]$ at the maximal ideal $\mathfrak{m}_P$. Infact, we have an exact sequence

$$0 \longrightarrow \mathfrak{m}_P \longrightarrow \mathcal{O}_{V,P} \longrightarrow \bar{k} \longrightarrow 0$$

which is given by $\frac{f}{g} \mapsto \frac{f(P)}{g(P)}$.

Definition 5.11. A smooth projective curve C over $\bar{k}$ is a projective variety of dimension 1, such that

$$\dim_k \left(\frac{\mathfrak{m}_P}{\mathfrak{m}_P^2} \right) = 1,$$

for all $p \in C$. Note that in this case, we have $\mathfrak{m}_P = (t)$ By Nakayama's Lemma. Any such t is called a uniformizer of C at P .

Definition 5.12. An affine Variety is said to be defined over k if its ideal has generators in the subring $k[x_1, \dots, x_n]$.

Definition 5.13. For a projective/affine variety, we can define $I(V/k)$ as the ideal of functions in $k[x_1, \dots, x_n]$ vanishing on V .

Remark 5.14. With the above notation, V is defined over k if and only if $I(V) = \bar{k}[x_1, \dots, x_n]I(V/k)$.

To finish the section we introduce the notion of morphisms defined over k . The idea is that we want to consider the class of morphisms which send k -rational points to k -rational points. We define this for projective varieties, and the general definition for quasi projective varieties follows from this.

Definition 5.15. A rational map $f: V \dashrightarrow W$ is said to be defined over k , if we may choose a representation:

$$f = [f_1(P) : \dots : f_m(P)],$$

where $f_i \in k(V)$.

Remark 5.16. Note that if f is defined over k and P is a k rational point of V , which is in the domain of f , then we have

$$\sigma(f(P)) = [\sigma(f_0(P)) : \dots : \sigma(f_m(P))] = [f_0^\sigma(\sigma(P)) : \dots : f_m^\sigma(\sigma(P))] = f(P).$$

Thus rational maps defined over k , send rational points to rational points. In fact, we will see in the exercises that this is the defining property of rational maps defined over k in the following sense; Suppose both W and V are defined over k , and let $\sigma \in G_k$ be any element of the Galois group. then σ acts on the set of rational maps from V to W if we define:

$$f^\sigma(P) = [f_0^\sigma(P) : \dots : f_m^\sigma(P)].$$

since $f_i^\sigma(P) = \sigma(f_i(\sigma^{-1}(P)))$, this definition really makes sense. We will then show that a rational map is defined over k if and only if $f^\sigma = f$ for all σ .

§5.2 Practice with non algebraically closed fields

Exercise. Exercise 1.4. Let $V/\mathbb{Q}$ be the variety

$$V : 5X^2 + 6XY + 2Y^2 = 2YZ + Z^2.$$

Prove that

$$V(\mathbb{Q}) = \emptyset.$$

Solution. Suppose there were rational solutions. By multiplying out a denominator we may assume that there is an integer solution and that the integer triple (X, Y, Z) has no common factor. Reducing mod 2, we see that

$$X \equiv Z \pmod{2},$$

Reducing the original equation mod 4, we have

$$0 \equiv 2Y^2 - 2Y(3X - Z) \pmod{4}.$$

since the second summand is zero mod 4, Y must be even. In particular, X and Y are odd since otherwise they have a gcd. But then reducing the original equation mod 8, we have

$$5 + 6XY \equiv 2YZ + 1 \Leftrightarrow 4 \equiv 2Y(3X - Z).$$

which is not possible as the right hand side is divisible by 8. $\square$

Exercise. Exercise 1.5. Let $V/\mathbb{Q}$ be the projective variety

$$V : Y^2 = X^3 + 17,$$

and let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be distinct points of V . Let L be the line through P_1 and P_2 .

(a) Show that

$$V \cap L = \{P_1, P_2, P_3\}$$

and express $P_3 = (x_3, y_3)$ in terms of P_1 and P_2 . (If L is tangent to V , then P_3 may equal P_1 or P_2 .)

(b) Calculate P_3 for

$$P_1 = (-1, 4) \quad \text{and} \quad P_2 = (2, 5).$$

(c) Show that if $P_1, P_2 \in V(\mathbb{Q})$, then $P_3 \in V(\mathbb{Q})$.

Solution. (a) The first part is a direct consequence of Bezout's theorem. For the second part let us assume, $x_2 \neq x_3$, since otherwise the third intersection point is the point at infinity. The line through P_1 and P_2 is then given by

$$Y = \frac{y_2 - y_1}{x_2 - x_1}(X - x_1) + y_1 =: mX + b$$

where $m = \frac{y_2 - y_1}{x_2 - x_1}$ and $b = y_1 - \frac{y_2 - y_1}{x_2 - x_1} \cdot x_1$. Since the third intersection point lies on V , we can plug this into the equation to receive

$$(mx + b)^2 = x^3 + 17 \Leftrightarrow x^3 - m^2x^2 - 2mbx + 17 - b^2 = 0$$

We know this has the three solutions x_1, x_2, x_3 , so we have

$$x^3 - m^2x^2 - 2bx + 17 - b^2 = (x - x_1)(x - x_2)(x - x_3).$$

Using Vietas relations, we see $x_1 + x_2 + x_3 = m^2$, thus $x_3 = m^2 - x_1 - x_2$. From this,

$$y_3 = mx + b = m(m^2 - x_1 - x_2) + b$$

(b) From (a), we know

$$x_3 = m^2 - x_1 - x_2 = \frac{1}{9} - 1 = -\frac{8}{9}.$$

and $y_3 = mx + b = \frac{109}{27}$. In total

$$P_3 = \left(-\frac{8}{9}, \frac{109}{27}\right).$$

(c) This follows immediately from (a), since the formula for P_3 , is a rational function in P_1 and P_2 with rational coefficients. □

Exercise. Exercise 1.8. Let $\mathbb{F}_q$ be a finite field with q elements and let

$$V \subset \mathbb{P}^n$$

be a variety defined over $\mathbb{F}_q$.

(a) Prove that the q^{th} -power map

$$\phi = [X_0^q, \dots, X_n^q]$$

is a morphism

$$\phi : V \longrightarrow V.$$

It is called the *Frobenius morphism*.

(b) Prove that ϕ is one-to-one and onto.

(c) If V is irreducible and has positive dimension, prove that ϕ is not an isomorphism.

(d) Prove that

$$V(\mathbb{F}_q) = \{P \in V : \phi(P) = P\}.$$

Solution. (a) This map is a morphism since it is given by homogenous polynomials of equal degree and they have no common vanishing.

(b) First we show that ϕ is one-to-one. Indeed, let $\phi(R) = \phi(S)$. Suppose wlog, that the the zero'th coordinate is nonzero, such that we then have:

$$[1 : (r_1/r_0)^q : \dots : (r_n/r_0)^q] = [1 : (s_1/s_0)^q : \dots : (s_n/s_0)^q].$$

Then we have for all i :

$$(r_i/r_0)^q = (s_i/r_0)^q \Leftrightarrow ((r_i/r_0) - s_i/r_0)^q \Leftrightarrow (r_i/r_0) = s_i/r_0$$

and so $R = S$. For subjectivity, we use that $\bar{k}$ is algebraically closed.

(c) If ϕ were an isomorphism, it would induce an isomorphism on function fields. However, the induced map on function fields is given by

$$\phi^* : \bar{k}(V) \rightarrow \bar{k}(V), t_i \mapsto t_i^q : \text{ where } t_i = \frac{X_i}{X_0}.$$

This can not be an isomorphism, otherwise for a smooth point it would induce an isomorphism of local rings and thus of differential modules at this point. However the induced differential map at the point is the zero map. The zero map is not an iso, unless X has dimension zero.

- (d) It is immediate that the points in $V(\mathbb{F}_q)$ are fixed points. On the otherhand, if P is a fixed point of the Frobenius morphism, then when we dehomogenize at some nonzero coordinate we see that the coordinates satisfy $t_i^q = t_i$. The polynomial $T^q - T$ has exactly q roots, all corresponding to the elements of $\mathbb{F}_q$. $\square$

Exercise. Exercise 1.9. If $m > n$, prove that there are no nonconstant morphisms

$$\mathbb{P}^m \longrightarrow \mathbb{P}^n.$$

Hint. Use the dimension theorem [111, I.7.2].

Solution. We don't follow the hint, but use the chance to refresh ourselves on linear systems. We know that maps from $\mathbb{P}^n \rightarrow \mathbb{P}^m$, correspond to basepoint linear systems of $\mathbb{P}^n$, of degree m . Hence, it suffices to show that such basepoint free linear systems do not exist. Assume δ is any linear system of degree m , and $\delta \subseteq |D|$. We know that $\text{Cl}(\mathbb{P}^n) \cong \mathbb{Z}$, and so $|D| \cong \mathcal{O}_{\mathbb{P}^n}(d)$ where the right hand side consists of hyper surfaces of degree $d := \dim |D|$. Thus we may choose a basis of δ , given by m homogeneous polynomials of degree d . However,

$$V_{\mathbb{P}^n}(F_1, \dots, F_m) \neq \emptyset,$$

since $m < n$. $\square$

Exercise. Exercise 1.10. For each prime $p \geq 3$, let

$$V_p \subset \mathbb{P}^2$$

be the variety given by the equation

$$V_p: X^2 + Y^2 = pZ^2.$$

- (a) Prove that V_p is isomorphic to $\mathbb{P}^1$ over $\mathbb{Q}$ if and only if

$$p \equiv 1 \pmod{4}.$$

- (b) Prove that for

$$p \equiv 3 \pmod{4},$$

no two of the V_p 's are isomorphic over $\mathbb{Q}$.

Solution. (a) If V_p is isomorphic over $\mathbb{Q}$ to $\mathbb{P}^1$, then it has $\mathbb{Q}$ -rational points. Multiplying out a denominator we get integer solutions as well and rescaling, we can assume that they don't share a common factor. Reducing mod p , we see that $X^2 \equiv -Y^2 \pmod{p}$. We can't have that p divides these terms however, since this would force p to divide Z as well. Hence, Y is a unit and we have

$$\left(\frac{-1}{p}\right) = 1,$$

where the above term denotes the Legendre symbol. However $\left(\frac{-1}{p}\right) = 1$ occurs iff $p = 2$ or $p \equiv 1 \pmod{4}$. For the converse, notice that if $p \equiv 1 \pmod{4}$, then $p = a^2 + b^2$, for two integers a, b . Thus V_p contains an affine rational point. Stereographic projection from this point to $\mathbb{P}^1$, then defines the desired isomorphism.

- (b) The solution involves applying Hensels lemma to obtain solutions in $\mathbb{Q}_p$ in V_q and vice versa. But this was slightly outside the scope of the course. $\square$

Exercise. Exercise 1.12.

- (a) Let V/K be an affine variety. Prove that

$$K[V] = \left\{ f \in \overline{K}[V] : f^\sigma = f \text{ for all } \sigma \in G_{\overline{K}/K} \right\}.$$

Hint. One inclusion is clear. For the other, choose some polynomial $F \in \overline{K}[X]$ with

$$F \equiv f \pmod{I(V)}.$$

Show that the map

$$G_{\overline{K}/K} \longrightarrow I(V), \quad \sigma \longmapsto F^\sigma - F$$

is a 1-cocycle; see (B §2). Now use (B.2.5a) to conclude that there exists a $G \in I(V)$ such that

$$F + G \in K[X].$$

- (b) Prove that

$$\mathbb{P}^n(K) = \left\{ P \in \mathbb{P}^n(\overline{K}) : P^\sigma = P \text{ for all } \sigma \in G_{\overline{K}/K} \right\}.$$

Hint. Write

$$P = [x_0, \dots, x_n].$$

If $P = P^\sigma$, then there is a $\lambda_\sigma \in \overline{K}^*$ such that

$$x_i^\sigma = \lambda_\sigma x_i \quad \text{for all } 0 \leq i \leq n.$$

Show that the map

$$\sigma \longmapsto \lambda_\sigma$$

gives a 1-cocycle from $G_{\overline{K}/K}$ to $\overline{K}^*$. Now use Hilbert's Theorem 90 (B.2.5b) to find an $\alpha \in \overline{K}^*$ such that

$$[\alpha x_0, \dots, \alpha x_n] \in \mathbb{P}^n(K).$$

- (c) Let

$$\phi : V_1 \longrightarrow V_2$$

be a rational map of projective varieties. Prove that ϕ is defined over K if and only if

$$\phi^\sigma = \phi$$

for every $\sigma \in G_{\overline{K}/K}$.

Hint. Use (a) and (b).

Solution. $\square$

§6 Lecture 5

We began this lecture by introducing divisors on curves.

Definition 6.1. Let C be a smooth projective curve defined over k . A **divisor** on C is a formal sum

$$D = \sum_{P \in C(\bar{k})} n_P P,$$

where $n_P \in \mathbb{Z}$ and $n_P = 0$ for all but finitely many points. The Free abelian points of all such points is denoted $Div(C)$. and the homomorphism

$$\deg : Div(C) \mapsto \mathbb{Z}, \sum_{P \in C(\bar{k})} n_P P \mapsto \sum_{P \in C(\bar{k})} n_P$$

is called the degree.

Since we have chosen C to be defined over k , the Galois group G_k acts on C . This action produces an action

$$G_k \times Div(C) \rightarrow Div(C), (\sigma, \sum_{P \in C(\bar{k})} n_P P) \mapsto \sum_{P \in C(\bar{k})} n_P \sigma(P).$$

Definition 6.2. A divisor $D \in Div(C)$ is said to be defined over k , if $\sigma(D) = D$ for all $\sigma \in G_k$.

Example 6.3

If $D = \sum_{P \in C(\bar{k})} n_P P$, is a divisor such that $n_P \neq 0$ only for the points in $c(k)$, then D is defined over k . However, this is not necessary. Indeed, consider $k = \mathbb{Q}$, and the divisor $D = 1 \cdot \sqrt{2} + 1 \cdot (-\sqrt{2}) \in Div(\mathbb{P}_k^1)$. Then $\sigma(D) = D$ for all σ , but this is not true for all coefficients.

Definition 6.4. A divisor D is called **effective** if all its coefficients are non negative and in this case we write $D \geq 0$. This defines a partial order on $Div(C)$ via

$$D_1 \geq D_2 : \Leftrightarrow D_1 - D_2 \geq 0$$

Recall that if C is a smooth curve and $p \in C$, then the local ring $\mathcal{O}_p := \mathcal{O}_{C,p}$ is a discrete valuation ring. It produces a well defined map:

$$\text{ord } \mathcal{O}_p \rightarrow \mathbb{N} \cup \{\infty\}, f \mapsto \sup\{n \in \mathbb{N} \mid f \in \mathfrak{m}_p^n\}.$$

Since we have $\bar{k}(C) = \text{Frac}(\mathcal{O}_p)$, this extends to a valuation on the function field:

$$\text{ord } \bar{k}(C) \rightarrow \mathbb{Z} \cup \{\infty\}, f/g \mapsto \text{ord } f - \text{ord } g$$

Notice also, that for $A \in \mathcal{O}_p$, we have

$$\text{ord } A = \infty \Leftrightarrow A \in \bigcap_{d \geq 0} \mathfrak{m}_p^d = \{0\}.$$

This map is a discrete valuation map:

Definition 6.5. A discrete valuation on a field F is surjective map,

$$\nu : F \rightarrow \mathbb{Z} \cup \{\infty\},$$

which satisfies:

- $\nu(xy) = \nu(x) + \nu(y)$
- $\nu(x + y) \geq \min\{\nu(x), \nu(y)\}$
- $\nu(x) = \infty \Leftrightarrow x = 0$.

for $f \in \bar{k}(C)$, we have $\text{ord}_p(f) \geq 0$ if and only if $f \in \mathcal{O}_p$. In this case we can evaluate f at p and $\text{ord}_p(f) = m$ if and only if f has an order m zero at p . Conversely if $\text{ord}_p(f) < 0$, then f has a pole there and we can prescribe $f(p) = \infty$.

Remark 6.6. From Algebraic geometry we know that a nonzero $f \in \bar{k}(C)$, that f has finitely many zeros and finitely many poles. Furthermore, since C is projective we know that if f has no poles, then f is constant.

Definition 6.7. If $f \in \bar{k}(C)^\times$ is a nonzero function, then we define

$$\text{div}(f) = \sum_{p \in C(\bar{k})} \text{ord}_p(f)P \in \text{Div}(C)$$

We will use as a black box

Proposition 6.8

for any $f \in \bar{k}(C)^\times$, we have $\deg(\text{div}(f)) = 0$.

Now suppose $f \in \bar{k}(C)^\times$ and $\sigma \in G_k$. Then we have

$$\begin{aligned} \text{div}(\sigma f) &= \sum_{p \in \bar{k}(C)} \text{ord}_p(\sigma(f))p \\ &= \sum_{p \in \bar{k}(C)} \text{ord}_p(f)\sigma(p) \\ &= \sigma(\text{div}(f)). \end{aligned}$$

Thus, for every $\sigma \in G_k$ we have a commutative diagram

$$\begin{array}{ccc} \bar{k}^\times & \xrightarrow{\text{div}} & \text{Div}^0(C) \\ \downarrow \sigma & & \downarrow \sigma \\ \bar{k}^\times & \xrightarrow{\text{div}} & \text{Div}^0(C) \end{array}$$

In particular, if f has coefficients in k , then $\text{div}(f)$ is defined over k .

Remark 6.9. Note that if $f \in \bar{k}$, then $\text{div}(f)$ captures all the information to know where f has zeros and poles and in what order they are.

Definition 6.10. Let $D \in \text{Div}(C)$ be a divisor. We define the $\bar{k}$ vector space $L(D)$ as:

$$L(D) = \{f \in \bar{k}(C)^\times \mid \text{div}(f) + D \geq 0\} \cup \{0\}.$$

The dimension of this space denotes $l(D)$.

Lemma 6.11

Let C be a smooth projective curve and $D \in \text{Div}(C)$. Then $l(D) < \infty$.

Proof. Note that if $D_1 \leq D_2$, and $f \in L(D_1)$, then $\text{div}(f) + D_2 \geq \text{div}(f) + D_1 \geq 0$. In particular $L(D_1) \subseteq L(D_2)$. Therefore, it suffices to prove that

$$l(D) < \infty \quad \forall D \geq 0.$$

In fact, we claim for all effective divisors

$$l(D) \leq \deg(D) + 1.$$

We will prove this by induction on $\deg D$. If $\deg D = 0$, then $D = 0$ since it was assumed to be effective. Thus, $f \in L(D)$ implies that f is regular on all of C and thus constant. So, $L(D) = k$, and $l(D) = 1 < \infty$. Suppose now D is an effective divisor of degree $\deg D > 0$ and that the claim is true for all divisors of degree $< \deg D$. Let P be any point that appears in D with a positive coefficient. We have an inclusion

$$L(D - P) \subseteq L(D).$$

If this is not a proper inclusion, we are done by induction. Otherwise there exists a function

$$f \in L(D) \setminus L(D - P).$$

if $n_p > 0$ is the coefficient in which P appears in D , we have $\text{ord}_P(f) = -n_p$. Choose a uniformizing parameter t of C at P . Then the map

$$L(D) \rightarrow k, x \mapsto (t^{n_p}x)(p),$$

is surjective and has kernel $L(D - P)$. Thus,

$$\dim L(D) = \dim L(D - P) + 1 \leq \deg D - 1 + 1 < \infty.$$

□

Remark 6.12. • Note that if $\deg D < 0$ then for any $f \in \bar{k}(C)^\times$, we have $\deg(\text{div}(f) + D) < 0$ and thus $L(D) = \{0\}$.

- for any $g \in \bar{k}(C)^\times$, we have

$$L(D) \cong L(D + \text{div}(g)),$$

where this isomorphism is induced via

$$\Phi_g: L(D) \rightarrow L(D + \text{div}(g)), f \mapsto fg.$$

The second remark above shows that $L(D)$ depends only on a certain quotient of the divisor group $\text{Div}(C)$.

Definition 6.13. The Picard group of C , denoted $\text{Pic}(C)$, is defined as the divisor group modulo function divisors:

$$\text{Pic}(C) := \text{coker}(\bar{k}(C)^\times \rightarrow \text{Div}(C), f \mapsto \text{div}(f))$$

§7 Lecture 6

In this lecture we aim to understand how the standard notions from algebraic curves, can be studied for non algebraically closed fields. We will assume for the rest of the lecture, that k is a perfect field, The First main lemma is the following

Lemma 7.1

Let D be a divisor defined over k , then $L(D)$ admits a basis of functions defined over k .

To prove this result, we will need some basic results from semi-linear algebra.

Lemma 7.2

Let V be a vector space $\bar{k}$ equipped with a continuous semi linear action of $G_k = \text{Gal}(\bar{k}/k)$. i.e:

- for all $v \in V$

$$\text{Stab}(v) = \{\sigma \in G_k \mid \sigma(v) = v\} \subseteq V,$$

has finite index.

- $\sigma(\lambda v) = \sigma(\lambda)\sigma(v)$ for all $\lambda \in \bar{k}$, $v \in V$

Then there is an isomorphism of $\bar{k}$ vector spaces:

$$V^{G_k} \otimes_k \bar{k} \xrightarrow{\cong} V, v \otimes \lambda \mapsto v\lambda.$$

Proof. The map above is clearly injective, so it suffice to prove it is surjective. That is to say that every vector in V can be written as a $\bar{k}$ -linear combination of vectors in V^{G_k} . Let $v \in V$ be any vector. By assumption its stabilizer H has finite index in the Galois group. Thus, from infinite Galois theory, this tell us that

$$[\bar{k}^H : k] < \infty$$

Thus, the normal closure L of $\bar{k}^H/k$ has finite index as well, and L/k is separable since k is perfect. Hence, we have constructed a finite galois extension. Let $\alpha_1, \dots, \alpha_n$ be a basis of L and $\text{Gal}(L/k) = \{\sigma_1, \dots, \sigma_n\}$. Consider now the n elements of V , defined as

$$w_j = \sum_{i=1}^n \sigma_i(\alpha_j)v = \sum_{i=1}^n \sigma_i(\alpha_j) \cdot \sigma_i(v), j = 1, \dots, n.$$

We claim $w_j \in V^{G_k}$ for all k . Indeed, let σ be any element of the galois group. Since L/k is normal, σ restricts to one of the σ_i , say σ_k . Then we have:

$$\sigma(w_j) = \sigma \left(\sum_{i=1}^n \sigma_i(\alpha_j) \sigma_i(v) \right) = \sum_{i=1}^n \sigma_i(\alpha_j) \sigma_i(v) = w_j,$$

where we used that for any group G and element $h \in G$, the map $G \xrightarrow{h} G$, is a bijection, and hence the last two terms differ by a permutation in summands. Now the defining

equations for w_j mean:

$$\begin{bmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_n(\alpha_n) \\ \vdots & \ddots & \vdots \\ \sigma_1(\alpha_n) & \cdots & \sigma_n(\alpha_n) \end{bmatrix} \cdot \begin{bmatrix} \sigma_1(v) \\ \vdots \\ \sigma_n(v_n) \end{bmatrix} = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$$

The matrix on the right hand side is invertible as its determinant is the discriminant of the field extension L/k . And so inverting this matrix we see that all the $\sigma_i(v)$ can be obtained as linear combinations of the w_i . In particular for the identity as well. $\square$

With this lemma proven, we can prove lemma 6.1.

Proof of 6.1. We Apply 6.2 on the following way; we will need

$$L(D) = L(D)_k \otimes_k \bar{k}.$$

For any $f \in L(D)$, we have $\sigma(f) \in L(\sigma(D)) = L(D)$, since we have assumed D to be defined over k . Hence, the Galois group defines an action on $L(D)$. In the exercise, we will show that it is continuous, and also semi-linear. $\square$

§7.1 Differentials

To better understand the spaces $L(D)$, we will need to introduce differentials. Indeed, the dimension of these spaces depends on them, Recall the definition of Kähler differentials.

Definition 7.3. Let $A \rightarrow B$ be a morphism of rings. The module of **Kähler differentials** $\Omega_{B/A}$ is the quotient of the free B -module F generated by symbols $db, b \in B$, by the submodule N which is generated by elements

- $d(b + b') - db - db'$ for $b, b' \in B$
- $d(bb') - bdb' - b'db$ for $b, b' \in B$
- da for $a \in A$.

Note that the module $\Omega_{B/A}$ is equipped with a map $B \rightarrow \Omega_{B/A}, b \mapsto db$. This map is A -linear and it is also a derivation. Infact, it is universal in the following sense:

Lemma 7.4 (Universal property of Kähler differentials)

For any A module M and any A -derivation $\delta : B \rightarrow M$, there exists a unique A -linear map $\tilde{\delta} : \Omega_{B/A} \rightarrow M$, such that the following diagram commutes.

$$\begin{array}{ccc} B & \xrightarrow{\delta} & M \\ \downarrow d & \searrow \tilde{\delta} & \uparrow \\ \Omega_{B/A} & & \end{array}$$

Furthermore, in the above sense, $\Omega_{B/A}$ is unique up to unique isomorphism.

Proof. The existence of the map $\tilde{\delta}$ is ensured by the homomorphism theorem. Uniqueness is also easy. $\square$

If M is a smooth manifold of dimension one, we know that at any point $p \in M$, its tangent space $T_p M$ is one dimensional and so its dual $\Omega_p(M)^1$ as well. Furthermore, if f is a smooth function defined locally near p , then df defines a basis of $\Omega_p(M)^1$. This holds also for the more general setting we are considering here.

Proposition 7.5

Let C be a smooth projective curve and consider $p \in C(\bar{k})$. Furthermore, let t be a local uniformizer at p . Then $\Omega_{\mathcal{O}_{C,p}/\bar{k}}$ is a free $\mathcal{O}_{C,p}$ module of rank one and it is generated by dt .

Proof. The proof was omitted; however, it can be found in my bachelor thesis: Algebraic curves and the Riemann-Roch theorem. $\square$

Definition 7.6. Let C be a smooth projective curve defined over k . We define the space of **meromorphic differential forms** on C to be the module

$$\Omega_C := \Omega_{\bar{k}(C)/\bar{k}}.$$

Similarly, the space

$$\Omega_C^{rat} := \Omega_{k(C)/k}$$

is called the space of **rational meromorphic differential forms** on C .

Remark 7.7. In the Exercises we will see that the module of Kähler differentials is compatible with localisations, meaning for any multiplicative subset $S \subseteq B$, we have isomorphism

$$\Omega_{B/A} \otimes_B S^{-1}B \cong S^{-1}\Omega_{B/A} \cong \Omega_{S^{-1}B/A}$$

Remark 7.8. Proposition 6.5 Shows us that Ω_C is a $\bar{k}$ vector space of dimension one.

Now suppose $\omega \in \Omega_C$ is a nonzero differential form. Choose a uniformizer t of C at $p \in C$. Then there exists a $g \in \bar{k}(C)$ such that $\omega = gdt$. We define $\text{ord}_p(\omega) := \text{ord}_p(g)$. This is easily checked to be independent of the choice of uniformizer.

Definition 7.9. For any $\omega \in \Omega_C \setminus \{0\}$ we define

$$\text{div}(\omega) = \sum_{P \in C(\bar{k})} (\text{ord}_P \omega) \cdot P$$

We will use as a fact that the support of a meromorphic differential form is finite.

Definition 7.10. The canonical divisor of C , denoted K_C , is the class in $\text{Pic}(C)$ of $\text{div}(\omega)$ for any nonzero meromorphic differential ω . It is well defined by Remark 6.8

Definition 7.11. Let C/k be a smooth projective curve. The genus of C is defined as

$$g(C) := l(K_C).$$

Remark 7.12. We say a differential form $\omega \in \Omega_C$ is regular if $\text{ord}_p(\omega) \geq 0$. Note that if $f \in L(K_C)$ nonzero, then we have

$$\text{div}(f) + \text{div}(\omega) = \text{div}(f\omega) \geq 0.$$

Thus, $f\omega$ is a regular differential form. Conversely, if $f\omega$ is regular, then $f \in L(K_C)$. Hence we have

$$L(K_C) \cong \{\omega \in \Omega_C \mid \omega \text{ is regular} \}$$

We can assume all the results about divisors and $L(D)$ spaces from algebraic geometry 1. In particular, we will make heavy use of the Riemann-Roch theorem later on. I will now only recall those results here that deal with $k \neq \bar{k}$.

Lemma 7.13

Suppose V is a smooth curve defined over k and $P \in \bar{k}$ is a k rational point. Then there exists a uniformizer t for P defined over k .

Proof. Since P is k rational, the galois group acts on the maximal ideal of the local ring at P . Choose uniformizer t defined over $\bar{k}$ first and then, consider $V = \text{span}_{\bar{k}}\{\sigma(t_i) : \sigma \in G_k\}$. Note again, that t_i has finite orbit, and so this is a finite dimensional vector space on which G_k acts continuously and semi linearly. By the lemma from the lecture V admits a basis defined over k . any of these elements from the basis then must have order one and are therefore uniformizers. $\square$

§8 Exercise Sheet 2

Exercise. Exercise 1.

Let F be a function field over k of one variable and X the associated smooth projective curve. Show the following facts:

- (1) For all $x \in X$, the residue field at x is k .
- (2) For all $f \in F \setminus k$, the extension

$$k(f) \subset F$$

is finite.

Solution. (1) We know that $\dim \mathcal{O}_x = \dim X$. Furthermore, since X is a smooth projective curve, we may choose a uniformizer t at x , such that $\mathfrak{m}_x = (t)$. By Krull's principle ideal theorem

$$\dim \mathcal{O}_x/(t) = \dim \mathcal{O}_x - 1 = \dim X - 1.$$

Thus the residue field has transcendence degree zero over k . Since k is algebraically closed, this implies that it is k .

- (2) If $f \in F \setminus k$, then f is transcendental over k , and thus, F is algebraic over $k(f)$. Since F is a finitely generated k algebra, $F/k(f)$ must then be a finite algebraic extension. $\square$

Exercise. Exercise 2.

Let

$$\pi : Y \longrightarrow X$$

be a nonconstant morphism between smooth projective curves over k .

For any divisor

$$D = \sum m_x x$$

on X , we set

$$\pi^*D := \sum_{\pi(y)=x} \text{ord}_y(\pi) m_x y,$$

where $\text{ord}_y(\pi)$ is the ramification index of π at y .

For any divisor

$$D = \sum m_y y$$

on Y , we set

$$\pi_*D := \sum_{x \in X} \left(\sum_{\pi(y)=x} m_y \right) x.$$

Show the following facts:

- (1) For any divisor D on X , we have

$$\deg(\pi^*D) = \deg \pi \cdot \deg D.$$

- (2) For any divisor D on Y , we have

$$\deg(\pi_*D) = \deg D.$$

- (3) Let f be a non-identically zero meromorphic function on X . Then

$$\pi^* \text{div}(f) = \text{div}(\pi^*f),$$

and deduce that

$$\deg(\text{div}(f)) = 0.$$

- (4) Let f be a non-identically zero meromorphic function on Y . Then

$$\pi_* \text{div}(f) = \text{div}(N(f)),$$

where $N(f)$ is the norm of f with respect to the extension

$$k(X) \subset k(Y).$$

Solution. (1) Since the pull back is additive it suffices to prove this for one point. But for one point, P , we have

$$\deg \pi^*P = \deg \left(\sum_{Q \mapsto P} \text{ord}_Q(\pi) \right) = \deg \pi$$

- (2) This follows again by linearity and seeing that the formula is true for a single point.

- (3) It is easy to see that the pullback behaves functorially. Thus we can compute:

$$\pi^* \text{div}(f) = \pi^*(f^*([0] - [\infty])) = (f \circ \pi)^*([0] - [\infty]) = \text{div}(f \circ \pi) = \text{div} \pi^*f.$$

The second claim then follows from

$$\deg \text{div}(f) = \deg(f^*([0] - [\infty])) = \deg f \cdot 0 = 0$$

(4)

□

Exercise. Exercise 3.

Let B be an algebra over a ring A . Show the following facts:

(1) If B is a localization or a quotient of A , we have

$$\Omega_{B/A}^1 = 0.$$

(2) For any A -algebra A' , we have

$$\Omega_{B'/A'}^1 \cong \Omega_{B/A}^1 \otimes_B B',$$

where

$$B' = B \otimes_A A'.$$

(3) Let $B \rightarrow C$ be a morphism of A -algebras. Then we have the exact sequence

$$\Omega_{B/A}^1 \otimes_B C \longrightarrow \Omega_{C/A}^1 \longrightarrow \Omega_{C/B}^1 \longrightarrow 0.$$

(4) For any multiplicative subset $S \subset B$, we have

$$S^{-1}\Omega_{B/A}^1 \cong \Omega_{S^{-1}B/A}^1.$$

(5) If

$$C = B/I$$

for an ideal $I \subset B$, we have an exact sequence

$$I/I^2 \longrightarrow \Omega_{B/A}^1 \otimes_B C \longrightarrow \Omega_{C/A}^1 \longrightarrow 0.$$

(6) If

$$B = A[x_1, \dots, x_n] \quad \text{and} \quad I = (F_1, \dots, F_r),$$

we have

$$\Omega_{C/A}^1 \cong \frac{C dx_1 \oplus \dots \oplus C dx_n}{(dF_1, \dots, dF_r)},$$

where

$$dF = \sum_{i=1}^n \frac{\partial F}{\partial x_i} dx_i.$$

(7) Let A be a local algebra over k with maximal ideal $\mathfrak{m}$ and residue field k . Then

$$\Omega_{A/k}^1 \otimes_A k \cong \mathfrak{m}/\mathfrak{m}^2.$$

(8) For a finite field extension $K \subset L$, we have

$$\Omega_{L/K}^1 = 0 \iff L \text{ is separable over } K.$$

(9) Let F be a function field over k . Then

$$\dim_F \Omega_{F/k}^1 = 1.$$

Solution. (1) We have

$$\begin{aligned} d(a/s) &= \frac{1}{s}da + ad\left(\frac{1}{s}\right) = 0 - a\frac{1}{s^2}ds = 0 \\ d(\bar{a}) &= d(a \cdot \bar{1}) = ad(\bar{1}) = 0 \end{aligned}$$

(2) Let us construct the isomorphism. We will do this using universal properties. First we construct a map

$$\Phi : \Omega_{B'/A'}^1 \longrightarrow \Omega_{B/A}^1 \otimes_B B'.$$

The natural choice is

$$\Phi(d(b \otimes a')) = db \otimes (1 \otimes a').$$

Let us show that such a map exists. Consider

$$\begin{array}{ccc} B \times A' & \xrightarrow{\varphi} & \Omega_{B/A}^1 \otimes_B B' \\ \downarrow & \nearrow \tilde{\varphi} & \\ B' & & \end{array}$$

where

$$\varphi(b, a') = db \otimes (1 \otimes a').$$

The map φ is A -balanced. Indeed, for $a \in A$,

$$\begin{aligned} \varphi(ab, a') &= d(ab) \otimes (1 \otimes a') \\ &= a db \otimes (1 \otimes a') \\ &= db \otimes (a \otimes a') \\ &= db \otimes (1 \otimes aa') \\ &= \varphi(b, aa'). \end{aligned}$$

Thus, by the universal property of the tensor product, it induces a map

$$\tilde{\varphi} : B' \longrightarrow \Omega_{B/A}^1 \otimes_B B'.$$

This map is an A' -derivation, and hence the universal property of Kähler differentials gives the desired map Φ .

Moreover, Φ is surjective, since

$$\Phi(d(b \otimes 1)) = db \otimes 1,$$

and the elements $db \otimes 1$ generate $\Omega_{B/A}^1 \otimes_B B'$ as a B' -module.

For injectivity, we construct an inverse. Consider the map

$$B \times B' \longrightarrow \Omega_{B'/A'}^1, \quad (b, b') \longmapsto b' d(b \otimes 1).$$

For every fixed $b' \in B'$, this is an A -derivation in the first factor. It therefore induces a map

$$\Omega_{B/A}^1 \times B' \longrightarrow \Omega_{B'/A'}^1, \quad (db, b') \longmapsto b' d(b \otimes 1).$$

This map is B -balanced. Indeed, for $c \in B$,

$$b' d(cb \otimes 1) = b' ((c \otimes 1)d(b \otimes 1) + (b \otimes 1)d(c \otimes 1)),$$

or, equivalently, writing

$$u : \Omega_{B/A}^1 \longrightarrow \Omega_{B'/A'}^1, \quad u(db) = d(b \otimes 1),$$

we have

$$b'u(c\omega) = b'(c \otimes 1)u(\omega) = ((c \otimes 1)b')u(\omega).$$

Consequently, the universal property of the tensor product gives a map

$$\Psi : \Omega_{B/A}^1 \otimes_B B' \longrightarrow \Omega_{B'/A'}^1, \quad db \otimes b' \longmapsto b' d(b \otimes 1).$$

It remains to check that the two maps are inverse. Since $d(1 \otimes a') = 0$ in $\Omega_{B'/A'}^1$, we have

$$\begin{aligned} \Psi\Phi(d(b \otimes a')) &= (1 \otimes a')d(b \otimes 1) \\ &= d((b \otimes 1)(1 \otimes a')) \\ &= d(b \otimes a'). \end{aligned}$$

Conversely,

$$\begin{aligned} \Phi\Psi(db \otimes b') &= \Phi(b' d(b \otimes 1)) \\ &= b'(db \otimes 1) \\ &= db \otimes b'. \end{aligned}$$

Therefore Φ and Ψ are inverse isomorphisms, and hence

$$\boxed{\Omega_{B'/A'}^1 \simeq \Omega_{B/A}^1 \otimes_B B'}.$$

- (3) We consider C to be an B algebra, and hence also an A algebra. First let us verify exactness at $\Omega_{C/A}$. The induced maps are

$$\begin{aligned} f : \Omega_{B/A} \otimes_B C &\rightarrow \Omega_{C/A}, \quad db \otimes c \mapsto c \cdot db \\ g : \Omega_{C/A} &\rightarrow \Omega_{C/B}, \quad dc \mapsto dc \end{aligned}$$

Notice that the last map is clearly surjective. It is even B linear since $db = 0$ in the $\Omega_{C/B}$. Furthermore,

$$(g \circ f)(db \otimes c) = g(c \cdot db) = c \cdot db = 0.$$

Hence it remains to show that every element in $\ker g$ is in the image of f . To show this, we can simply show that g induces an iso $f \text{ coker } f \cong \Omega_{C/B}$ has the universal property of Kähler differentials of C over B . Indeed, we have a B derivation

$$C \mapsto (\Omega_{C/A})/(\Omega_{B/A} \otimes_B C), \quad dc \mapsto \overline{dc}.$$

this map is B linear. By the universal property of Kähler differentials, this induces a map $h : \Omega_{C/B}$, which is exactly the inverse of $\bar{g}$.

- (4) Let $C = S^{-1}B$ and apply (3) and (1) to obtain a surjective map

$$\Omega_{B/A}^1 \otimes_B S^{-1}B \longrightarrow \Omega_{S^{-1}B/A}^1 \longrightarrow 0$$

This map is injective because one can construct an inverse:

$$\Omega_{S^{-1}B/A}^1 \longrightarrow \Omega_{B/A}^1 \otimes_B S^{-1}B, \quad d\left(\frac{b}{s}\right) \mapsto \frac{1}{s}db,$$

which one easily verifies to be well defined.

(5) By (1) we have that $\Omega_{C/B} = 0$. Hence, by (3), we get an exact sequence:

$$\Omega_{B/A}^1 \otimes_B C \longrightarrow \Omega_{C/A}^1 \longrightarrow 0.$$

We also have a map: $I \rightarrow \Omega_{B/A} \otimes_B C, x \mapsto dx \otimes 1$, This map vanishes on I^2 , since

$$d(xy) \otimes 1 = xdy \otimes 1 + ydx \otimes 1 = dy \otimes x + dx \otimes y = 0 + 0.$$

Thus we have a map

$$I/I^2 \rightarrow \Omega_{B/A} \otimes_B C, \bar{x} \mapsto dx \otimes 1.$$

It is clearly that the composition

$$I/I^2 \rightarrow \Omega_{B/A} \otimes_B C \rightarrow \Omega_{C/A},$$

vanishes. we must show that this induces an isomorphism

$$(\Omega_{B/A} \otimes_B C)/(I/I^2) \rightarrow \Omega_{C/A},$$

again the inverse is guaranteed by the universal property of the kähler differentials as in the last task.

(6) By the previous task, we have

$$\Omega_{C/A} \cong \frac{\Omega_{B/A} \otimes_B C}{(I/I^2)}$$

First note, that $\Omega_{B/A} = Bdx_1 \oplus \cdots \oplus Bdx_n$ and so tensoring with C , we get that the numerator is exactly as desired. Under that map $I/I^2 \rightarrow \Omega_{B/A} \otimes_B C$ from the last task, we see that the image is precisely $dF_1, \dots, dF_r$ as desired.

(7)

$$\Omega_{A/k} \otimes_A (A/\mathfrak{m}) \cong \frac{\Omega_{A/k}}{\mathfrak{m}\Omega_{A/k}},$$

It is clear, that $\Omega_{A/k} \cong \mathfrak{m}$, since the differentials of things not in $\mathfrak{m}$, vanish

□

§9 Elliptic curves

We now have the tools in place to talk about the main subject of this course. To simplify our discussion, we will fix a perfect field k and an algebraic closure $\bar{k}$

Definition 9.1. An **elliptic curve** over k , is a smooth genus one curve over k , along with a choice of k rational point $O \in E$.

Remark 9.2. By convention we mean projective curve when we say curve.

Example 9.3

We have class of examples given by

$$E_\lambda : y^2 = x(x-1)(x-\lambda) \subseteq \mathbb{P}^2,$$

for $\lambda \in k \setminus \{0, 1\}$ (note we mean the projective closure of this curve). They are smooth by the Jacobian criterion and genus one by the degree formula.

Example 9.4

A non example over $\mathbb{Q}$ is given by the projective closure of

$$E : y^2 = x^3,$$

which is singular at the origin, and infact it is genus 0. A more subtle nonexample is

$$E : 3x^3 + 4y^3 + 5 = 0,$$

which is a smooth genus one curve over $\mathbb{Q}$, but it has no $\mathbb{Q}$ rational point.

The reasons elliptic curves are so interesting are that they are the first example of an abelian variety. They admit a natural group structure. This group structure is obtained from the Picard group. As we will see later, it also has a nice geometric interpretation in the case of plane curves.

Proposition 9.5

Let E be an elliptic curve over k , then the map

$$AJ : E(\bar{k}) \rightarrow \text{Pic}(X)^0(E), P \mapsto [P] - [O],$$

is a G_k equivariant bijection. That is to say, it is a bijection which satisfies:

$$\sigma(AJ(P)) = AJ(\sigma(P)), \forall \sigma \in G_k.$$

The map AJ , is called the Abel-Jacobi map.

Proof. The fact that this map is G_k equivariant is essential trivial, so we show only that it is a bijection.

Injectivity: Assume we have two point P, Q with $AJ(P) = AJ(Q)$. This means that

$$[P] - [O] \sim [Q] - [O] \Leftrightarrow \exists f : [P] - [Q] = \text{div } f.$$

However, if $P \neq Q$, then f defines a degree 1 map to $\mathbb{P}^1$, and hence an isomorphism. This contradicts that E has genus one, and so $P = Q$.

Surjectivity: Let $[D]$ be any class in $\text{Pic}^0(E)$. We want a point P and a function f with

$$D = [P] - [O] + \text{div } f.$$

For this consider the divisor $D + O$, it has degree 1, and hence by the Riemann-Roch we have:

$$l(D + O) - l(K_E - D - O) = 1 + 1 - 1 = 1$$

on the otherhand $\deg(K_E) = 2g - 2 = 0$, and so the second term in the above formula vanishes. Hence there exists a nonzero function $f \in L(D + O)$. By definition, this means:

$$D' := \text{div } f + D + O \geq 0,$$

Since $\deg D' = 1$, and $D' \geq 0$, we have that $D' = P$ for some $P \in E$, and the claim follows.

□

Using the above bijection, we can pull back the group structure from $\text{Pic}^0(E)$ to E to define a group law on E . Under this bijection, O gets mapped to the neutral element. At this point, it is not yet clear if this group law behaves well with the algebraic structure on E . Later, we will see that this is the case, i.e addition and inversion define morphisms on E . To prove this, and many more facts it will be useful to have some sort of standard model for our elliptic curves. As it turns out, we can embed them over k into $\mathbb{P}^2$, and this embedding is essentially unique, i.e unique up to one isomorphism.

Proposition 9.6

Let (E, O) be an elliptic curve over k , then there exists $x, y \in k(E)^\times$ non constant, such that

$$\Phi : E \rightarrow \mathbb{P}^2, P \mapsto [x(P) : y(P) : 1],$$

defines an isomorphism to a curve:

$$E' : Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3,$$

with $a_i \in k$ and $\Phi(O) = [0 : 1 : 0]$, and $x = \frac{X}{Z}$, $y = \frac{Y}{Z}$.

Proof. By the riemann roch theorem, we have

$$l(n[O]) = n, \forall n \geq 1.$$

Notice that all these divisors are defined over k . For $n = 1$, we get the constants, which have basis 1. for $n = 2$, we can extend this to a basis $\{1, x\}$ where x may be chosen to be defined over k . For $n = 3$, we can extend this to a basis $\{1, x, y\}$ where y is also defined over k . Consider now the elements

$$\{1, x, y, xy, x^2, x^3, y^2\} \in L(6[O])^{G_k}.$$

Since the latter has dimension 6 over k , they satisfy a nontrivial combination

$$\lambda_0 y^2 + \lambda_1 xy + \lambda_2 y = \lambda_3 x^3 + \lambda_4 x^2 + \lambda_5 x + \lambda_6, \text{ with } \lambda_i \in k.$$

Notice that if $\lambda_0 = 0$, then λ_3 must be zero as well in order for orders at O to match. The following this, we see that all other coefficients must be zero, in order for orders to match. Hence $\lambda_0 \neq 0 \neq \lambda_3$. If we multiply the equation by λ_0 and replace y by $\lambda_0 y$, This produces:

$$y^2 + \lambda_1 xy + \lambda_2 y = \lambda_3 x^3 + \lambda_4 x^2 + \lambda_5 x + \lambda_6, \text{ for new } \lambda_i \in k.$$

If we multiply this equation by λ_3^2 , replace x by $\lambda_3 x$ and y by $\lambda_3 y$, this becomes

$$y^2 + \lambda_1 xy + \lambda_2 y = x^3 + \lambda_4 x^2 + \lambda_5 x + \lambda_6, \text{ for new } \lambda_i \in k.$$

Which is an equation of the desired form. Thus, the map

$$\Phi : E \rightarrow \mathbb{P}^2, \Phi = [x : y : 1],$$

maps to a curve in Weierstraß form. This map is clearly non constant, and hence surjective. To conclude the proof, we need to show only, that Φ is degree one. Let C denote the curve to which Φ , maps, the pullback of Φ has image:

$$\Phi^* \bar{k}(C) = \bar{k}(x, y).$$

Thus we need to show that $[\bar{k}(E) : \bar{k}(x, y)] = 1$. On the otherhand, we have

$$\begin{aligned} [\bar{k}(E) : \bar{k}(x, y)] \cdot [\bar{k}(x, y) : \bar{k}(y)] &= [\bar{k}(E) : \bar{k}(y)] \\ [\bar{k}(E) : \bar{k}(x, y)] \cdot [\bar{k}(x, y) : \bar{k}(x)] &= [\bar{k}(E) : \bar{k}(x)] \end{aligned}$$

Consider the morphism represented by :

$$\varphi_x = E \rightarrow \mathbb{P}^1, P \mapsto [x(P) : 1].$$

We have that $\deg \varphi_x = \deg \varphi_x^*[\infty] = 2$, Since x only has one pole, namely at O and of order 2. Thus $[\bar{k}(E) : \bar{k}(x)] = 2$. Ther exact same computation shows that $[\bar{k}(E) : \bar{k}(y)] = 3$, and thus $[\bar{k}(E) : \bar{k}(x, y)]$ is a divisor of 2 and 3, which leaves $[\bar{k}(E) : \bar{k}(x, y)] = 1$ as the only possibility. $\square$

The above proposition tells us that to study elliptic curves, it suffices to study curves in **Weierstrass** form. Meaning that E is given as a plane curve solving a cubic equation as above (monic in y^2 and x^3 and such that O is the point at infinity. The next proposition tells us, that the Weierstraß forms are almost unique.

Proposition 9.7

Let (E, O) be an elliptic curve and suppose we have two Weierstraß forms for E :

$$\begin{aligned} E_1 : y^2 + a_1xy + a_3y &= x^3 + a_2x^2 + a_4x + a_6 \\ E_2 : y^2 + b_1xy + b_3y &= x^3 + b_2x^2 + b_4x + b_6 \end{aligned}$$

Then E_2 is obtained from E_1 via a change of variables

$$x' = u^2x + 2, \quad y' = u^3y + u^2sx + t, \quad \text{with } u, s, t \in k \text{ and } u \neq 0.$$

Proof. Assume there exists an isomorphism $\psi = [x' : y' : 1]$ between E_1 and E_2 , with $\infty \mapsto \infty$. Notice that x and x' both have a order 2 pole at O and so we have two basis

$$B_1 = \{1, x\}, B_2 = \{1, x'\},$$

of $L(2[O])$. The basis transformation matrix must look like:

$$\begin{pmatrix} 1 & v \\ 0 & u \end{pmatrix},$$

so $x' = ux + v$, and $u \neq 0$. Similarly for we have two basis $\{1, x, y\}$ and $\{1, x', y'\}$ of $L(3[O])$, and the basis transformation matrix must look like:

$$\begin{pmatrix} 1 & v & \alpha \\ 0 & u & \beta \\ 0 & 0 & \gamma \end{pmatrix},$$

Substitute this into the equation of E_2 to recieve:

$$(\alpha + \beta x + \gamma y)^2 + b_1(\alpha + \beta x + \gamma y)(ux + v) + b_3(\alpha + \beta x + \gamma y) = (ux + v)^3 + b_2(ux + v)^2 + b_4(ux + v) + b_6$$

Since this equation needs to be a scalar multiple of the defining equation of E_2 , we see that $\gamma^2 = u^3$. Let $u' = \frac{\gamma}{u}$ and the result follows. $\square$

Remark 9.8. For most cases, we can do better than the above representation, i.e achieve a more simple form. Assume we have a curve in Weierstraß form given by

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6,$$

First assume that $\text{char } k \neq 2$. Then we may complete the square on the righthand side to obtain:

$$(y + \frac{1}{4}a_1x + \frac{1}{4}a_3)^2 = x^3 + b_2x^2 + b_4x + b_6$$

Hence the change of coordinates $y' = (y + \frac{1}{4}a_1x + \frac{1}{4}a_3)$ changes this into an equation of the form:

$$y'^2 = x^3 + b_2x^2 + b_4x + b_6.$$

If $\text{char } k \neq 3$ as well, we may complete the cube on the RHS to obtain a form

$$y'^2 = (x + \frac{1}{3}b_2)^3 - 27(x + \frac{1}{3}b_2)c_4 - 56c_6,$$

hence, the change of variables $x' = (x + \frac{1}{3}b_2)$ achieves a form:

$$y'^2 = x'^3 + c_4x' + c_6.$$

Now that we have established weierstraß forms for our elliptic curves, we can associate a special differential, called the invariant differential.

Definition 9.9. Let E be an elliptic curve with Weierstraß equation:

$$E : Y^2Z + a_1YXZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3.$$

let $y := \frac{Y}{Z}$ and $x := \frac{X}{Z}$. The **invariant differential** of E is defined as

$$\omega = \frac{dx}{2y + a_1x + a_3} = \frac{dy}{3x^2 + 2a_2x + a_4 - a_1y}$$

First we should take a moment to see that the two equations of ω agree. But this is easy; let f denote the equation of E dehomogenized at Z . Differentiating f , we see:

$$f_x dx + f_y dy = 0 \Leftrightarrow \frac{dx}{f_y} = -\frac{dy}{f_x}.$$

From this presentation, we can conclude that $\text{div}(\omega) = 0$. Indeed, for any affine point $P = (x_0, y_0) \in E$. we see that note both f_y and f_x can be zero since E is smooth. Consider the tangent line to E at P , it is given by $f_x(P)(x - x(P)) + f_y(P)(y - y(P)) = 0$. If f_y is nonzero, this is not the line $x - x(P) = 0$, and so $x - x(P)$, is a uniformizer of E at P . From the first representation of ω , we see $\text{ord}_P(\omega) = 0$. The same reasoning holds for $f_x \neq 0$, if we choose the second representation. Finally, since $\deg \text{div}(\omega) = 2g - 2 = 0$, ω must have order 0 at ∞ as well.

§10 The group Law

In the last section we have established a G_k equivariant bijection

$$E(\bar{k}) \longleftrightarrow \text{Pic}^0(E), P \mapsto [P] - [O].$$

Using this map, we have established a group structure on the $E(\bar{k})$. We would like to investigate this structure more closely. In particular, we will show that the group operations are morphisms.

Theorem 10.1

Let E/k be an elliptic curve, then addition defines a morphism:

$$E \times E \rightarrow E, (P, Q) \mapsto P + Q.$$

To prove this we will establish the following first:

Proposition 10.2

Let E be an elliptic curve in Weierstraß form, and P, Q, R three points on E . Then $P + Q + R = 0$ if and only if there exists a line $L \subseteq \mathbb{P}^2$, such that $L \cap E = \{P, Q, R\}$

Proof. Assume first that $P + Q + R = 0$, using the group isomorphism to $\text{Pic}^0(E)$, this means that

$$[P] + [Q] + [R] - 3[O] \sim 0.$$

and thus there exists a rational function f with

$$\text{div}(f) = [P] + [Q] + [R] - 3[O].$$

This f is clearly in $L(3[O])$, and since this space has basis, $1, x, y$, we may write

$$f = ax + by + c = \frac{aX + bY + cZ}{Z},$$

Consider now the line $L : X + bY + cZ = 0$, it must intersect E , in $\{P, R, Q\}$. Conversely, suppose there exists a line L , with $L \cap E(\bar{k}) = \{P, Q, R\}$. Let the equation of L be $aX + bY + cZ = 0$ and set $f = ax + by + c$. Then we see $\text{div}(f) = \{P, Q, R\} - 3[O]$, as desired. $\square$

Now let us consider negation.

Proposition 10.3

The map

$$E(\bar{k}) \rightarrow E(\bar{k}), P \mapsto -P,$$

is an isomorphism of projective varieties.

Proof. Let E be in Weierstraß form:

$$E : Y^2Z + a_1YXZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3$$

. Since E is a smooth projective curve, it suffices to prove that $P \mapsto -P$, defines a morphism on the affine part of E , since rational maps are defined in codimension one for normal varieties. Notice also that if P is affine, then $-P$ must be as well. For two affine points P and R , we have

$$P + R = 0 \Leftrightarrow P + Q + O = 0 \text{ Notice } O = 0.$$

From the previous lemma this occurs if and only if P, R, O lie on a line. A line given by $aX + bY + cZ$ contains $[0 : 1 : 0]$ if and only if $b = 0$. Hence, the affine representation of L (after rescaling) is $x + c = 0$. Since $P = (x_0, y_0)$ needs to be on this line, we have that

the line is given by $L : x - x_0 = 0$. Thus if R is $-P$, it has the same x coordinate as P . Plugging this into the affine Weierstraß form, we get:

$$y^2 + a_1yx_0 + a_3y - x_0^3 - a_2x_0^2 - a_4x_0 - a_6 = 0 = (y - y_0)(y - y')$$

Using vietes relations, the second colution for y satisfies

$$y' = -(a_1x_0 + a_3 + y_0)$$

Hence the map $P \mapsto -P$ is represented in the affine by

$$(x, y) \mapsto (x, -(a_1x + a_3 + y)),$$

Which is clearly a morphism since it is polynomial. Notice that this morphism is defined over k . $\square$

Remark 10.4. The above two lemmas give us a geometric interpretation for plane curves: to find the sdum of two points, connect them with a line, and find the third intersection point of the line with the curve. Then, connect this third point with a line through O and find the corresponding third intersection point. In practice, this gives a much better way of calculating in the group.

Our last preperation for the main theorem is that translating by a point is also a morphism.

Proposition 10.5

Let Q be a point in E , then the map

$$\tau_Q : E \rightarrow E, P \mapsto P + Q,$$

is an isomorphism of projective varieties. If Q is defined over k , then this morphism is defined over k as well.

Proof. The last part follows simply from the fact that the bijection to the Picard group is G_k equivariant. This is also clearly a bijection with inverse τ_{-Q} , and so we only need to prove this is a morphism. This is easily seen by the combination of the last two propositions. First, for $P \in E$, we have that the point R' such that P, Q, R lie on a line is clearly given as a rational function in the coordinates of Q and R . Then, the map $R \mapsto -R$ is also a morphism by the previous proposition. Combining the two propositions, we get the result. With this in case we can now prove the result. $\square$

Proof of Theorem 9.1. First we show that addition restricted to $W := E \times E \setminus (\Delta \cup E \times \{O\} \cup \{O\} \times E)$ is a morphism. Indeed this is essentially obvious since the third point is given again by coordinates in the first two points. Hence we need to show that this rational map is defined on the diagonal as well. Fix a point $Q \neq O$. Then the morphism may be represented as:

$$W \xrightarrow{(P,P) \mapsto (P,P+Q)} (E \times E) \setminus W \xrightarrow{(R,S) \mapsto R+S} E \xrightarrow{U \mapsto U-Q} E.$$

$\square$

§11 Isogenies

In the last few sections we introduced elliptic curves and investigated some basic properties. In particular, we saw that they carry a natural group structure given by the Picard group. We ask ourselves the question: what is the right notion of morphism for elliptic curves. Clearly we want something stronger than the just being a morphism of algebraic curves, since ideally our morphisms would respect the group structure. For this to hold, it is necessary, that the morphism

$$\varphi : E \rightarrow E',$$

maps $O \rightarrow O'$. In fact, much later, we will see that this is sufficient as well.

Definition 11.1. Let (E, O) and (E', O') be two elliptic curves. An **Isogeny** between E and E' is a morphism $\varphi : E \rightarrow E'$, which maps $O \mapsto O'$. Two elliptic curves are called **isogeneous**, if there exists a nonconstant isogeny from one to the other (in which case we shall see that there also exists an isogeny in the other direction).

Remark 11.2. Notice that by the theorem of the dimension of fibers, any non constant isogeny

$$\varphi : E \mapsto E',$$

induces a finite field extension

$$k(E)/k(E'), \text{ with } [k(E)/k(E')] = \deg \varphi.$$

Hence we may talk about the degree of a non constant isogeny. By convention we also define the degree of the zero isogeny to be zero, such that the degree of an isogeny is multiplicative under composition.

Let us denote by $\text{Hom}(E_1, E_2)$ the set of isogenies from E_1 to E_2 . Since E_1 and E_2 are abelian groups, and their group structures are morphisms, $\text{Hom}(E_1, E_2)$ is again an abelian group, by setting

$$(\Phi + \Psi)(P) := \Phi(P) + \Psi(P), \text{ and } (-\Phi)(P) = -\Phi(P).$$

If $E_1 = E_2$, we can denote this simply by $\text{End}(E_1)$. For this case, we have a little more structure, we can define a multiplication of two isogenies, as their compositions. At this point it is not clear that this works distributively. We shall prove this later. $\sim$

Remark 11.3. for $\text{End}(E_1)$, we can define

$$[m] : E \rightarrow E, p \mapsto mp = [P] + \cdots + [P],$$

for $m \geq 0$ and $[m] = -[-m]$ for $m < 0$. Notice that if E is defined over K , then these morphism are defined over k .

Proposition 11.4

let $m \in \mathbb{Z}$ be a nonzero integer, then the map $[m]$ is nonconstant. Furthermore, if E_1, E_2 are two elliptic curves, then $\text{Hom}(E_1, E_2)$ is a torsion free $\mathbb{Z}$ module. Lastly, for any elliptic curve E , the endomorphism ring $\text{End}(E_1)$ is ring of characteristic zero with no zero divisors.

Proof. The last statement is an immediate consequence of the first two and the fact that if $\psi \circ \phi = [0]$, then either $\psi = 0$, or $\Phi = 0$, since in any other case, both maps are surjective. Furthermore, the second statement follows from the first. since $m\Phi = [m] \circ \Phi$, and so if Φ is nonzero, then the last composition will be if the first statement is true. Finally, for the first statement it suffices to prove the result for $m > 0$. This will follow in three steps. First consider the case $m = 2$. Consider the Weierstraß form of E . Then we have for affine points $P \in E$, that $2P = [O]$ if and only if $2P + O = O$, which in turn occurs if and only if the tangent line of E through P is $x - x(P)$. Thus, if f is the defining equation of f , these are those points for which $f_x(P) = 0$. If $\text{char } k \neq 3$, then we immediately see that this happens at finitely many places. If $\text{char } k = 3$, then in particular, $\text{char } k \neq 2$, and E has a weierstraß equation of the form

$$y^2 = x^3 + a_2x^2 + a_4x + a_6,$$

If either $a_2 \neq 0$ or $a_4 \neq 0$, we are again done for the same reason as before. If both are zero, the equation is $y^2 = x^3 + a_6$. Since $\text{char } k = 3$, this equation is $y^2 = (x + \sqrt[3]{a_6})^3$, which is singular, contradiction. Hence in all cases, $[2] \neq [0]$. from this it easily follows that $[2^n] \neq [0]$ by induction. Hence to complete the proof we need to show the result only for odd integers. Let us first specialize the case to where $\text{char}(k) \neq 2$. Then we have a Weierstraß equation as above. And we may find a nontrivial point P , with $2P = O$, simply by choosing its x . coordinate to be a root of $x^3 + a_2x^2 + a_4x + a_6$, and its y coordinate to be zero. For this point we have:

$$[m](P) = [2r + 1](P) = [r][2](P) + P = P.$$

Thus P is a point not mapped to O , under $[m]$, and therefore $[m]$ is nonconstant. For $\text{char } k = 2$ this proof fails! The curve may fail to have points of 2 torsion. However in this case we can by an analagous, yet slightly messier computation show that the curve has 3 torsion points, and write $m = 3q + r$. with $r \in \{1, -1\}$. $\square$

Definition 11.5. Suppose E is an elliptic curve. The **m torsion subgroup** of E , denoted $E[m]$, is defined as the set of points $P \in E$ for which $mP = O$. The **torsion** subgroup of E , denoted by E_{tors} is defined as

$$E_{\text{tors}} := \bigcup_{m>0} E[m].$$

Similarly, we can define the m torsion points defined over k , and the torsion subgroup defined over k ,

We are now ready to prove the main theorem, namely that isogenies are homomorphisms.

Theorem 11.6

Let

$$\Phi : E_1 \rightarrow E_2,$$

be an isogeny of elliptic curves. Then Φ is a group homomorphism.

Proof. If Φ is the constant isogeny there is nothing to prove, so let us assume wlog that Φ is nonconstant. Thus, Φ is a finite map (and hence also quasi finite). In particular we

may define a pushforward map

$$\Phi_* : \text{Pic}^0(E_1) \rightarrow \text{Pic}^0(E_2), \left[\sum_{P \in E} n_P P \right] \mapsto \left[\sum_{P \in E} n_P \phi(P) \right].$$

Note that if $D = D' + \text{div}(f)$, then from sheet two we know:

$$\Phi_*(D) = \Phi_*(D') + \Phi_*(\text{div}(f)) = \Phi_*(D') + \text{div}(N(f)).$$

This pushforward is clearly a group homomorphism. Let $\kappa_i : E_i \rightarrow \text{Pic}^1(E_1)$, denote the canonical isomorphisms. Then we have:

$$\kappa_2(\Phi(P)) = [\Phi(P)] - [O_2] = [\Phi(P)] - [\Phi(O_1)] = \Phi_*(\kappa_1(P)).$$

In particular $\Phi = \kappa_2^{-1} \circ \Phi_* \circ \kappa_1$ is a composition of group morphisms. $\square$

Corollary 11.7

Let $\Phi : E_1 \rightarrow E_2$ be a nonconstant isogeny. Then $\ker \Phi$ is a finite subgroup of E_1 .

Proof. $\ker \Phi$ is the preimage of a closed set and hence closed. Since we assume it not to be all of E , it consists of finitely many points. Infact, we have:

$$\sum_{P \mapsto O_2} e_P(\phi) = \deg \Phi,$$

and thus there are at most $\deg \phi$ many points in the kernel with equality if and only if Φ is unramified. $\square$

Theorem 11.8

Let $\Phi : E_1 \rightarrow E_2$ be a nonconstant isogeny.

1. For all points $Q \in E_2$, $\Phi^{-1}(Q)$ consists of exactly $\deg_s \Phi$ many points and for every $P \in E_2$, we have $e_P(\Phi) = \deg_i(\phi)$.
2. The map

$$\Psi : \ker \Phi \rightarrow \text{Gal}(\bar{k}(E_1)/\Phi^*(\bar{k}(E_2))), P \mapsto \tau_P^*,$$

is well defined and an isomorphism.

Proof. 1. The idea is that since E is a group, most notions are rather symmetric. Indeed we know that the formula in (1) holds for almost all points. If Q is a point for which it does not hold, choose a point Q' for which it does hold. Since Φ is surjective, the point $Q - Q'$ is in the image, say $\Phi(R)$. If we translate E_1 by R , we obtain a bijection of $\Phi^{-1}(Q_1)$ and $\Phi^{-1}(Q_2)$. Let $P, P' \in E_1$ be two points mapping to the same point under Φ . Let $R = P_1 - P_2$. Then, we have $\Phi(R) = O$, and thus the map $\Phi \circ \tau_R$ is an isogeny. Since τ_R is an isomorphism, we have:

$$e_\Phi(P_1) = e_{\Phi \circ \tau_R}(P_1) = e_\Phi(\tau_R(P_1)) \cdot e_{\tau_R}(P_1) = e_\Phi(\tau_R(P)) = e_\Phi(P_2).$$

Thus, using the usual formula for the degree of the pullback, the claim follows.

2. First suppose $P \in \ker \Phi$, and consider the isomorphisms τ_P^* . If $f = \Phi^*(g) \in \Phi^*(\bar{k}(E_2))$. Then we have:

$$\tau_P^*(f)(Q) = f(P + Q) = \Phi^*(g)(P + Q) = g(\Phi(P) + \Phi(Q)) = g(\Phi(Q)) = f(Q).$$

Thus the map is well defined. Furthermore, this map is also clearly a group homomorphism. By (1), we also know, that $|\ker \phi| = \deg_s \phi$. From field theory, we also have that

$$\text{Gal}(\bar{k}(E_1)/\Phi^*(\bar{k}(E_2))) \leq \deg_s \Phi.$$

Thus, it suffice to show that Ψ is injective. But this is easy, since for $\tau_P^* = \text{Id}$ to happen, every function on must take the same value for Q and $Q + P$ for all Q . for $P \neq O$, these are different functions, and using the Riemann-Roch we can always separate points with functions.

□

Corollary 11.9

If $\Phi : E_1 \rightarrow E_2$ is a separable isogeny, then $\ker \Phi = \deg \Phi$, and every element in E_2 has exactly $\deg \Phi$ many preimages. Furthermore Φ^* induces a galois extension.

Proof. Everything except the last claim is immediate. For the last claim we can use that an extension is galois if and only if its galois group has the same size as the field extension, which is guaranteed from the previous lemma. □

As for most abelian categories, we also have a homomorphism theorem for elliptic curves as we shall see in the following. However, we do need to restrict ourselves to separable morphisms, in order for this to work.

Corollary 11.10 (Homomorphism theorem)

Let $\phi : E_1 \rightarrow E_2$ and $\psi : E_1 \rightarrow E_3$ be two non constant isogenies and assume ϕ is separable. if $\ker \phi \subseteq \ker \psi$, then there exists a unique isogeny $\chi : E_2 \rightarrow E_3$, such that the following diagram commutes:

$$\begin{array}{ccc} E_1 & \xrightarrow{\psi} & E_3 \\ \downarrow \phi & \nearrow \chi & \\ E_2 & & \end{array}$$

Proof. The uniqueness of this map already follows from the group theoretic version of this theorem. The crux lies in showing that this map is also an isogeny. Since we have ϕ is sperable, the extension:

$$\bar{k}(E_1)/\phi^*(\bar{k}(E_2)),$$

is a galois extension. Furthermore, since $\ker \phi \subseteq \ker \psi$, we have from tge previous theorem, that

$$\text{Gal}(\bar{k}(E_1)/\phi^*(\bar{k}(E_2))) \subseteq \text{Gal}(\bar{k}(E_1)/\psi^*(\bar{k}(E_3))).$$

This means that every automorphism of $\bar{k}(E_1)$, which fixes $\phi^*\bar{k}(E_2)$, also fixes $\psi^*\bar{k}(E_3)$. From The fundamental theory of galois theory, this must force:

$$\psi^*\bar{k}(E_3) \subseteq \phi^*\bar{k}(E_2),$$

This inclusion is precisely the pullback of the desired isogeny (here we use the equivalence of categories to function fields). $\square$

As we have seen, the kernels of nontrivial isogenies are finite groups. The question arises: is every finite subgroup of E also the kernel of some isogeny?

Theorem 11.11

Let E be an elliptic curve and $G \subseteq E$ be a finite subgroup. Then there exists a unique elliptic curve E' and a separable isogeny $\Phi: E \rightarrow E'$, such that $\ker \phi = G$

Proof. For each $P \in G$, we get an automorphism τ_P^* of $\bar{k}(E)$, denote by $(\bar{k}(E))^G$, the fixed field. From basic galois theory, we know from Artins theorem that the extension:

$$(\bar{k}(E))/(\bar{k}(E))^G$$

is Galois with Galois group G . in particular this extension is separable. furthermore, the field $(\bar{k}(E))^G$ is of transcendent degree 1 over k and thus it corresponds to the function field of a unique smooth projective curve C . The inclusion $\bar{k}(C) \hookrightarrow \bar{k}(E)$ is then induced by a unique (finite) separable surjective morphism

$$\psi: E \rightarrow C.$$

We claim that ψ is unramified, which is to say that every point in C has exactly $\deg \psi$ many preimages. Notice, that for every function $f \in \bar{k}(C)$, and every point $Q \in E$, and every $P \in G$, we have

$$f(\psi(Q + P)) = (\psi^* f)(Q + P) = (\tau_P^*)(\psi^* f)(Q) = (\psi^* f)(Q) = f(\psi(Q)),$$

which shows that Q and $Q + P$ must map to the same point. In particular, every element in C has at least $|G|$ many preimages; Choose one, and take all the G translations. On the otherhand, we always have at most this many preimages from which the claim follows. Finally, Applying The hurwitz formula, we have

$$2g(E) - 2 = \deg \phi(2g(C) - 2),$$

which only works for $g(C) = 1 = \deg \phi$. Φ becomes an isogeny if we take $\phi(O)$ to be our designated point. $\square$

§12 The Invariant differential

Recall that for an elliptic curve E in Weierstraß form:

$$E: y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6,$$

we defined the invariant differential ω as

$$\omega = \frac{dx}{2y + a_1x + a_3} = \frac{dy}{3x^2 + 2a_2x + a_4 + a_1y}.$$

As we have seen, it is holomorphic. with degree 0 everywhere. We now aim to justify its name. It comes from the fact that it is invariant under translations:

Proposition 12.1

For every $P \in E$, we have

$$\tau_P^* \omega = \omega.$$

Proof. Let $P \in E$ be any point and consider τ_P^* . Since Ω_E is one dimensional, and $\omega \neq 0$, we there exists a nonzero function field element f_P such that

$$\tau_P^* \omega = f_P \omega.$$

Since τ_P is an isomorphism, f_P is nonzero and $\tau_P^* \omega$ must have order 0 everywhere (uniformizers are mapped to uniformizers). Thus, for any Q , we have

$$0 = \text{ord}_Q(f_P \omega) = \text{ord}_Q(f_P) + \text{ord}_Q(\omega) = \text{ord}_Q(f_P),$$

This implies that f_P is a constant. Consider now the map

$$E \mapsto \mathbb{P}^1, P \mapsto [f_P : 1].$$

Taking for granted that this is a morphism, we can see that it must be constant since $[0 : 1]$ is not in the image. Since for $P = O$, this is $[1 : 1]$ the claim follows. To see that τ_Q is a morphism, one computes this directly in coordinates, which is a bit messy. $\square$

Theorem 12.2

Suppose $\phi, \psi \in \text{Hom}(E_1, E_2) \setminus \{0\}$, and ω is the invariant differential of E_1 . Then we have

$$(\phi + \psi)^* \omega = \psi^* \omega + \phi^* \omega.$$

Proof. The proof is lengthy and difficult, so I have omitted it. The main idea is a direct computation. $\square$

Assuming the above theorem, we derive many interesting consequences:

Corollary 12.3

For the invariant differential ω on E , we have for all $m \in \mathbb{Z}$.

$$[m]^* \omega = m\omega$$

For $m = 0$, this is clear. For $m < 0$ appeal to the previous theorem and the $m = 0$ case to reduce this to $m > 0$. For $m = 1$, the statement is also obvious. For $m > 1$, use induction and the previous theorem to conclude:

$$[m + 1]^* \omega = ([m] + [1])^* \omega = (m + 1)\omega.$$

Using this fact, we can then provide a nice proof that multiplication by m , if $\text{char } k$ does not divide m , yields a finite separable morphism.

Proof. By the above we have

$$[m]^* \omega = m\omega,$$

thus, it is impossible for $[m]$ to be constant. Furthermore, we know that the extension is separable if and only if the induced map on differentials is injective. $\square$

Remark 12.4. This is a very strong tool we have used: a function field extension of curves is separable if and only if the induced maps on differentials is injective

§13 The Dual isogeny

We come to the final section of the lecture. When we introduced isogenies, we said that two elliptic curves are isogeneous, if there exists a nonconstant isogeny from one curve to the other. There is an immediate question that arises: is this relation symmetric? The answer to this is positive, however to prove it, we will need to prove that if $\phi : E \rightarrow E'$ is a nonconstant isogeny, then there also exists an isogeny in the other direction. Of course, if it exists, there are infinitely many such maps, since we may concatenate them with any multiplication map. There is however a canonical choice called the **dual isogeny**. The construction is natural: Pass from E and E' to their Picard zero groups $\text{Pic}^0(E)$ and $\text{Pic}^0(E')$ via the group isomorphism κ_1 and κ_2 . Then, ϕ induces a pullback of divisors ϕ^* , which then restricts to a map

$$\phi^* : \text{Pic}^0(E') \rightarrow \text{Pic}^0(E).$$

. then the dual map should be given as

$$\hat{\phi} : E' \rightarrow E, P \mapsto (\kappa_1^{-1} \circ \phi^* \circ \kappa_2)(P).$$

While this map is clearly a group homomorphism. it is not clear at all that it is an isogeny. As it will turn out, this is especially subtle if ϕ is not separable.

When Pondering over choices for $\hat{\phi}$, one may also try another approach: if ϕ has degree m , then choose for any $P \in E'$ a preimage $Q \in E$, and declare, $\hat{\phi}(P) = mQ$. As it will turn out, this definition is equivalent to the one given, but this will take some work. It is, in particular, independent of choice of Q .

Theorem 13.1

Let $\phi : E_1 \rightarrow E_2$ be a non constant isogeny, of degree m .

- (a) There exists a unique isogeny $\hat{\phi} : E_2 \rightarrow E_1$, such that

$$\hat{\phi} \circ \phi = [m].$$

- (b) As a group homomorphism, $\hat{\phi}$ is given by: Interpreting $P - O$ as a degree zero divisor, pulling it back to a divisor via ϕ^* and summing up the points via the group structure in E .

Remark 13.2. Notice that the description in (b) is not quite what we have introduced as the canonical choice above, but we will show that these agree.

Proof of Theorem 12.1. First we show uniqueness. Suppose $\hat{\phi}_1, \hat{\phi}_2$ are two maps with the property. Then we have:

$$0 = \hat{\phi}_1 \circ \phi - \hat{\phi}_2 \circ \phi = (\phi_1 - \phi_2) \circ \phi.$$

Since $\text{End}(E_1)$ has no zero divisors, and the second factor is nonzero by assumption, this forces $\phi_1 = \phi_2$.

Next, we will show that, given existence, the association $\phi \mapsto \hat{\phi}$ is functorial (Hence we have constructed a contravariant functor). Suppose $\psi: E_2 \rightarrow E_3$ is another isogeny of degree n . Then we compute:

$$(\hat{\phi} \circ \hat{\psi}) \circ (\psi \circ \phi)(P) = (\hat{\phi} \circ (\hat{\psi} \circ \psi) \circ \phi)(P) = (\hat{\phi} \circ [n] \circ \phi)(P) = (\hat{\phi} \circ \phi) \circ [n](P) = [nm](P).$$

By uniqueness, this implies $\widehat{\psi \circ \phi} = \hat{\phi} \circ \hat{\psi}$. This will be the main tool in helping us separate into two cases:

1. ϕ is separable
2. ϕ is a Frobenius morphism

Indeed, as is shown in the section on the Frobenius morphism (TODO!!!) any ϕ can be factored as a composition as two such morphisms. Let us consider first the case, where ϕ is separable. This occurs if and only if $|\ker \phi| = m$, which by Lagrange's theorem implies that every element of $\ker \phi$ has an order dividing m . Thus $\ker \phi \subseteq \ker [m]$. By the Homomorphism theorem, this immediately implies the result. Now the hard case: ϕ is the Frobenius morphism. In this case, if ϕ is obtained by applying the p -Frobenius morphism, multiple times, and thus we can assume, wlog, that ϕ is the p -th Frobenius morphism.

Consider now the multiplication by $[p]$ map. Since its pullback on differentials is the zero map, this isogeny is not separable, and thus it factors as:

$$[p] = \psi \circ \phi^e,$$

for some $e \geq 1$. Define the map $\hat{\phi}$ as:

$$\hat{\phi} := \psi \circ \phi^{e-1},$$

which implies the result. To finish the proof we need to show (b); i.e we need to compute the image of a point under $\hat{\phi}$.

$$\begin{aligned} \text{sum}(\phi^*([P] - [O])) &= \sum_{Q \mapsto P} e_Q(\phi)Q - \sum_{Q' \mapsto O} e_{Q'}(\phi)Q' \\ &= \deg_i \phi \left(\sum_{Q \mapsto P} Q - \sum_{Q' \mapsto O} Q \right) \end{aligned}$$

Let us investigate the inner sum. Choose a specific Q with $Q \mapsto P$, then for any other $\tilde{Q}$, there is a unique Q' mapping to O , for which $Q = \tilde{Q} - Q'$. Thus, rearranging the above sum, we see that it equates to the Q times the number of such points, which we know to be $|\ker \phi|$. Hence the last sum equates to:

$$\deg_i \phi \left(\sum_{Q \mapsto P} Q - \sum_{Q' \mapsto O} Q \right) = [\deg_i \phi] \circ [|\ker \phi|](Q) = [\deg \phi](Q).$$

Via this description, it also becomes clear that the constructed map is exactly the desired map. $\square$

Remark 13.3. By convention we define the Dual isogeny of $[0]$ to be again the zero map.

Let us now uncover the most important properties of The Dual map.

Proposition 13.4 (Properties of the Dual Isogeny)

Let $\phi : E \rightarrow E'$ be an isogeny. The Dual satisfies the following properties:

(a) Let $m = \deg \phi$, then

$$\hat{\phi} \circ \phi = [m]$$

(b) $\widehat{\phi \circ \psi} = \hat{\psi} \circ \hat{\phi}$.

(c) $\widehat{\phi + \psi} = \hat{\phi} + \hat{\psi}$

(d) $\widehat{[m]} = [m]$, and $\deg[m] = m^2$

(e) $\deg \phi = \deg \hat{\phi}$.

(f) $\hat{\hat{\phi}} = \phi$.

Proof. Everything is trivial if any of the appearing isogenies are trivial. Hence Assume they are all non constant. Furthermore, We have already shown (b) In the proof of Theorem 12.1. Let us prove the remaining parts:

(a) . For a point $P \in E_2$, let us compute $\phi \circ \hat{\phi}(P)$. Since ϕ , is nontrivial, we can write $P = \phi(Q)$, and then find:

$$\phi \circ \hat{\phi}(P) = \phi \circ \hat{\phi}(\phi(Q)) = \phi(mQ) = [m]\phi(Q) = [m](P).$$

(c) Blackbox. Can be done in coordinates.

(d) We prove the first part by induction. It is true for $m = [0]$ and $m = [1]$. Then, using the black box in (C) we have

$$\widehat{[m+1]} = \widehat{[m]} + \widehat{[1]} = [m] + [1] = [m+1].$$

By the definition of the dual we then have

$$[\deg[m]] = [m] \circ [m] = [m^2],$$

which implies the result, since End is torsion free.

(e) Let $d = \deg \phi$, we then have:

$$d^2 = \deg[m] = \deg(\hat{\phi} \circ \phi) = \deg \hat{\phi} \cdot \deg \phi = \deg \hat{\phi} \cdot d,$$

from which the claim follows.

(f) This follows from (a) and (e).

□

We will now see the power of the Dual isogeny: it allows us to understand the multiplication maps in more detail

Theorem 13.5

Let E be an elliptic curve and $m \in \mathbb{Z} \setminus \{0\}$, then we have:

1. $\deg[m] = m^2$
2. If m is coprime to $\text{char } k$, then

$$E[m] := \ker[m] \cong \frac{\mathbb{Z}}{m\mathbb{Z}} \times \frac{\mathbb{Z}}{m\mathbb{Z}}.$$

3. If $\text{char } k = p$, then either

$$E[p^e] = \{O\}, \text{ for all } e \geq 1,$$

$$E[p^e] = \frac{\mathbb{Z}}{p^e\mathbb{Z}}, \text{ for all } e \geq 1,$$

Proof. 1. This was already shown.

2. If m is coprime to $\text{char } k$, then $[m]$ is separable, and $\deg[m] = |\ker[m]|$, which implies that $E[m]$ is an abelian group of order m^2 . Now for each prime q dividing m , let $q^{\nu(q)}$, be the maximal power of q dividing m . For which we have:

$$E[m] \cong \prod_{q|m} E[q^{\nu(q)}].$$

Hence, it suffices to prove this for prime powers. Hence assume $m = q^e$, $p \neq q$. The Classification of finite abelian groups then implies a unique factorisation:

$$E[m] \cong \prod_{i=1}^r \mathbb{Z}/q^{t_i}\mathbb{Z},$$

No consider the number of elements of order $\{0, q\}$. On the one hand, we know this is $|E[q]| = q^2$. On the other hand, given the above factorization, these are the elements which have order $\in \{0, q\}$ in all factors. Hence there are q^r such factors, which implies $r = 2$. Thus:

$$E[m] \cong \left(\frac{\mathbb{Z}}{q^t\mathbb{Z}}\right)^2$$

3. Assume now that $m = p^e$. Let ϕ be the p -th Frobenius morphism. then we have

$$|E[m]| = \deg_s([p^e]) = \deg_s([\hat{\phi} \circ \phi])^e = \deg_s([\hat{\phi}])^e,$$

Since the Frobenius morphism is purely inseparable. We also have

$$\deg \hat{\phi} = \deg \phi = p,$$

There are two cases: $\hat{\phi}$ is purely inseparable, in which case, the separability degree is zero, and $|E[m]| = 1$, or it is separable, in which case $|E[m]| = p^e$. For $e = 1$, this implies the result. for $e = 2$, then use the fact that it must contain a point of order p^2 again, and so on.

□